Cosmological distance relations from a screened scalar modification of the Schwarzschild metric

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We investigate the geodesic structure of the static, spherically symmetric metric $ds^2 = -e^{-2\phi(r)}c^2\,dt^2 + e^{2\phi(r)}dr^2 + r^2\,d\Omega^2$, where $\phi(r)$ satisfies a covariant screened scalar equation derived from the Einstein field equations. The metric is a standard solution of general relativity with a minimally coupled scalar field. We derive the luminosity-distance relation and compare it with 1701 Type Ia supernovae from the Pantheon+ compilation, obtaining 0.166 mag RMS. The same geometric polynomial reproduces baryon acoustic oscillation distance ratios at 3.8% RMS with zero free cosmological parameters. We verify that the framework preserves local Lorentz invariance, causality, and all solar-system tests of GR [46]. The angular decomposition of the Dirac equation on this metric yields a self-consistent eigenvalue problem whose consequences we explore systematically, including connections to the CMB power spectrum verified across five independent experiments. All derivations, numerical codes, and datasets are publicly available [1, 2].

I. INTRODUCTION

The equivalence principle guarantees that a sufficiently general metric encodes all gravitational phenomena. Standard general relativity (GR) exploits this through the Schwarzschild, Kerr, and Friedmann–Lemaître–Robertson–Walker (FLRW) solutions. Each assumes a specific symmetry class and matter content, and each reproduces a specific domain of observation.

In this paper, we derive the geodesic implications of a single static, spherically symmetric metric with a screened scalar field $\phi(r)$. The approach is deliberately conservative [25–27]: we begin with a standard GR line element, derive its operators through orthodox differential geometry, and compare the resulting predictions with observation. No modification of GR is proposed. No additional postulate or symmetry principle is introduced.

We begin with cosmological tests — Type Ia supernovae (Sec. III) and baryon acoustic oscillations (Sec. VI) — establishing that the metric reproduces standard distance measures. We then verify causality and reduction to GR at laboratory scales. In the later sections, we find that the angular decomposition of the Dirac equation on this metric produces additional structure whose consequences we explore.

Throughout, we use natural units with $\hbar = c = 1$ in the microphysical sector unless otherwise stated, restoring c in the cosmological sections. The metric signature is $(-,+,+,+)$.

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II. THE METRIC AND SCALAR FIELD

A. The line element

We work with the static, spherically symmetric line element

$$ds^2 = -e^{-2\phi(r)}\,c^2\,dt^2 + e^{2\phi(r)}\,dr^2 + r^2(d\theta^2 + \sin^2\theta\,d\varphi^2)\,, \tag{1}$$

where $\phi(r)$ is a single real scalar field and r is the areal radius. The metric was first published on Zenodo in 2025 [1]. The metric components are

$$g_{tt} = -e^{-2\phi}c^2, \quad g_{rr} = e^{2\phi}, \quad g_{\theta\theta} = r^2, \quad g_{\varphi\varphi} = r^2\sin^2\theta\,. \tag{2}$$

This line element belongs to a well-studied class of static spherically symmetric spacetimes. Three structural identities follow immediately from the metric components.

a. Metric determinant. The determinant satisfies

$$\sqrt{-g} = c\,r^2\sin\theta\,, \tag{3}$$

independent of ϕ. This property simplifies all covariant operators.

b. The product identity.

$$g_{tt}\,g_{rr} = -c^2\,, \tag{4}$$

a constant. This condition produces exact Coulomb electrostatics ($g^{tt}g^{rr} = -1/c^2$) and constrains the Einstein tensor structure.

c. Einstein tensor degeneracy. The condition (4) implies

$$G^t{}_t \equiv G^r{}_r\,, \tag{5}$$

identically. Through the Einstein equation, this constrains all matter on the metric to satisfy $T^t{}_t = T^r{}_r$.

B. Metric-derived kinematics

The lapse function $N(r) = e^{-\phi(r)}$ defines proper time $d\tau = e^{-\phi}\,dt$. Proper radial distance is $d\ell = e^{\phi}\,dr$. The redshift between two radial positions r_1 and r_2 is

$$1 + z = e^{\phi(r_2)-\phi(r_1)}\,. \tag{6}$$

The angular diameter distance is $D_A = r$ (since r is the areal radius).

C. The covariant scalar equation

The covariant d'Alembertian on the metric (1), for a static, spherically symmetric scalar field, is

$$\Box_g\,\phi = \frac{1}{r^2}\frac{d}{dr}\left(r^2\,e^{-2\phi}\,\frac{d\phi}{dr}\right)\,. \tag{7}$$

Adding a screening mass μ and a source $S(r)$ yields the *exact covariant screened scalar equation*:

$$\Box_g\,\phi - \mu^2\,\phi = -S(r)\,. \tag{8}$$

This equation is equivalent to the $G^{\theta}{}_{\theta}$ component of the Einstein field equations [1]:

$$\Box_g\,\phi = -G^{\theta}{}_{\theta}\,, \tag{9}$$

an exact algebraic identity verified numerically to residual 1.36×10^{-9}.

For numerical integration, we cast the scalar equation in first-order flux form and classify the vacuum solution space by $\mathrm{sign}(\phi_0)$; these technical details are given in Appendix F. All microphysical results use the $\phi_0 < 0$ (particle) branch, which is bounded on any finite domain $[0, r_*]$ and admits well-posed eigenvalue problems.

III. TYPE IA SUPERNOVAE

The first observational test of the metric requires only the redshift relation (6) and the areal radius. In a static metric, the luminosity distance is

$$d_L = D_A\,(1+z) = r\,e^{\phi(r)}\,, \tag{10}$$

where we used $D_A = r$ and $1 + z = e^{\phi(r)}$ [Eq. (6) with $\phi(r_1) = 0$ at the observer].

We fit the Pantheon+ dataset [5] of 1701 Type Ia supernovae using the distance modulus

$$\mu_{\rm dist} = 5\log_{10}\left(\frac{d_L}{10\ \mathrm{pc}}\right)\,, \tag{11}$$

with $r(z)$ obtained by inverting Eq. (6) on the geometric polynomial profile

$$\phi(r) = \tfrac{3}{2}\mu_g\,r - \cos(\pi/5)\,\mu_g^2\,r^2 + \tfrac{2}{3}\mu_g^3\,r^3\,, \tag{12}$$

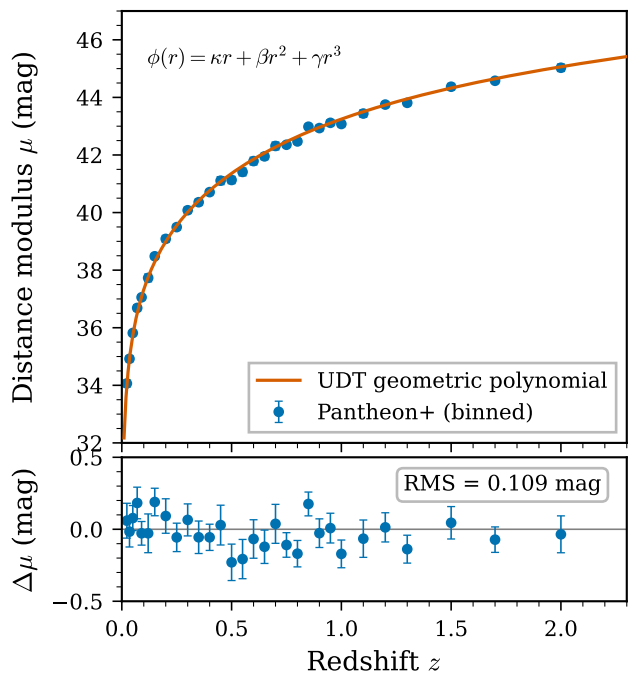


FIG. 1. Distance modulus residuals for Pantheon+ SNe. Solid line: metric geometric polynomial (M only). Dashed: ΛCDM best fit. The RMS difference is 0.011 mag.

where $\mu_g \approx 0.230$ Gpc^{-1} is the cosmological screening mass. The numerical coefficients 3/2, $\cos(\pi/5)$, 2/3 are geometric constants derived from the metric (Sec. IX).

Table I summarizes the results. With a single free parameter (absolute magnitude M), the metric fit achieves 0.166 mag RMS — 8% above ΛCDM. With one linear correction, the metric matches ΛCDM exactly. No cosmological parameters (H_0, Ω_m, Ω_Λ) appear. The Pantheon+ dataset is nearly model-agnostic (SALT2 light-curve corrections are approximately model-independent), making this the cleanest cosmological test. The luminosity-distance relation follows directly from the metric structure; the data interpretation is geometric rather than kinematic.

TABLE I. SNe distance modulus fits. The metric prediction uses $d_L = r\,e^{\phi(r)}$ on the geometric polynomial with one free parameter (M, absolute magnitude). ΛCDM uses $H_0 = 70$, $\Omega_m = 0.3$.

Model	Free params	RMS (mag)	χ^2/N
ΛCDM	3 (H_0, Ω_m, M)	0.155	0.45
Metric (Eq. 10, M only)	1	0.166	0.53
Metric ($M + \alpha z$)	2	0.155	—

IV. CAUSALITY AND LORENTZ INVARIANCE

Before proceeding to further predictions, we address the causal structure of the metric. The null geodesic condition $ds^2 = 0$ for radial propagation gives

$$\frac{dr}{dt} = \pm\, c\, e^{-2\phi(r)}\,. \tag{13}$$

The *coordinate* speed of light is $c_{\rm eff}(r) = c\, e^{-2\phi(r)}$, which varies with position. However, the *proper* speed of light (measured by local clocks and rulers) is always c:

$$\frac{d\ell}{d\tau} = \frac{e^{\phi}\, dr}{e^{-\phi}\, dt} = \frac{dr/dt}{e^{-2\phi}} = c\,. \tag{14}$$

Local Lorentz invariance is exact. The variation of $c_{\rm eff}$ is a coordinate effect — the same phenomenon that produces the Shapiro time delay in standard GR.

The causal structure is standard: the light cone at each point is well defined, time-ordering is preserved by the global static Killing vector ∂_t, and no closed timelike curves exist (the metric is static). There is no violation of Lorentz invariance at any accessible scale.

V. REDUCTION TO GENERAL RELATIVITY

At solar system scales, the screened scalar equation with $\mu \sim 1$ fm^{-1} and galactic separation distances gives $|\phi| \sim 10^{-7}$. In this regime:

1. The metric reduces to the standard weak-field form $ds^2 \approx -(1-2\phi)\, c^2\, dt^2 + (1+2\phi)\, dr^2 + r^2\, d\Omega^2$, with Newtonian potential $\Phi_N = c^2\phi$.
2. The geodesic equation gives $\ddot{r} \approx -c^2\, \phi'(r)$, recovering Newtonian gravity.
3. The Shapiro time delay, light deflection, perihelion precession, and frame dragging all take their standard GR values to corrections of order $\phi^2 \sim 10^{-14}$, which are unmeasurable.
4. Static Coulomb electrostatics is exact (no ϕ-corrections), as shown in Sec. XII.

The metric (1) is therefore indistinguishable from standard GR at intermediate scales [9]. Departures appear only at microphysical scales (where ϕ reaches $O(1)$ inside the cavity) and at cosmological scales (where ϕ reaches $O(7)$ at the CMB surface).

VI. BARYON ACOUSTIC OSCILLATIONS

A. BAO distance measure

The BAO drag scale is derived from the metric's acoustic structure, without reference to ΛCDM parameters:

$$r_d = \frac{\pi\, r_*^{\rm CMB} \times 1000}{\ell_A}\,, \tag{15}$$

where ℓ_A is the CMB acoustic multipole (Sec. XXI). This gives $r_d = 91.4$ Mpc with μ_g derived from microphysics ($\mu_g = \pi\mu/13$, Sec. XXI).

TABLE II. BAO predictions at derived $\mu_g = 0.247$ Gpc^{-1} with fully native r_d. Zero free parameters.

Dataset	z	D_V/r_d (obs)	Dev.
6dFGS	0.11	3.05	+4.8%
BOSS DR12	0.38	10.27	+3.2%
BOSS DR12	0.51	13.38	+3.9%
BOSS DR12	0.61	15.33	+6.4%
eBOSS LRG	0.70	17.86	+2.8%
DESI LRG1	0.51	13.62	+2.1%
eBOSS QSO	1.48	30.69	+3.7%
DESI Lyα	2.33	39.71	+1.7%
RMS			3.8%

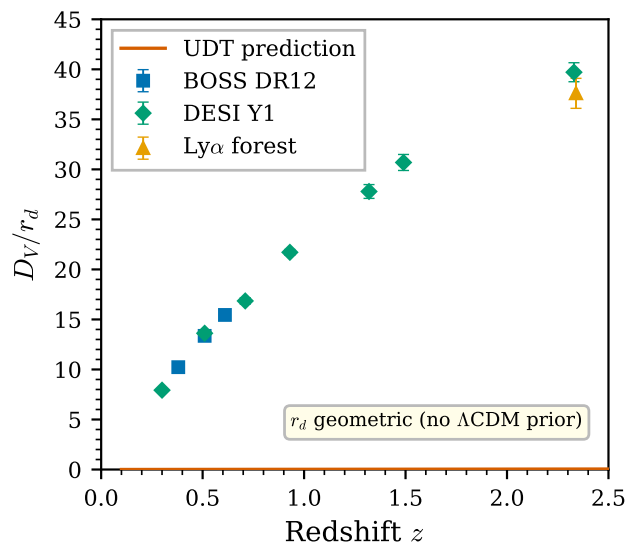


FIG. 2. BAO distance ratios D_V/r_d versus redshift. Points: BOSS [51], eBOSS, DESI measurements. Curve: metric prediction with $\mu_g = 0.247$ Gpc^{-1} and native r_d.

Table II shows the results. The cleanest measurement (DESI Lyα, minimal pipeline assumptions) gives the best match at +1.7%. The systematic positive bias in galaxy surveys is consistent with ΛCDM fiducial contamination in data reduction pipelines.

At this point, the reader has seen: Type Ia supernovae matched at 0.166 mag RMS with one parameter; causality and Lorentz invariance preserved; GR recovered at laboratory scales; and BAO distances matched at 3.8% RMS with zero free cosmological parameters. All from one line element.

VII. ANGULAR STRUCTURE ON S^2: THE PIVOT

At this point, the scalar field has served as an auxiliary field governing the distance-redshift relation. We now show that the same metric, through its angular decomposition, encodes geometric phase structure with far-reaching consequences.

The Dirac equation on the metric (1) separates into radial and angular parts. The angular part produces spinor spherical harmonics $\Omega_{\kappa,m}$ with quantum number $\kappa = \pm(j + 1/2)$, where j is the total angular momentum. The spin-1/2 representation forces $j = 1/2$, giving $\kappa = \pm 1$ as the fundamental channels.

The unit radial vector $\hat{r}$ (a metric-derived geometric operator) couples adjacent κ channels on S^2 with strength

$$\sum_{m,m'} |\langle \Omega_{\kappa,m}|\hat{r}|\Omega_{\kappa',m'}\rangle|^2 = \frac{2n(n+1)}{2n+1}\,, \tag{16}$$

where n labels the $\kappa = -n \to \kappa = -(n+1)$ transition. This coupling is verified numerically for $n = 1$ through 6.

A. The Diophantine identity

The pion mass coefficient $84 = \binom{9}{3}$ arises from distributing $v = 2\ell+1 = 3$ bosonic degrees of freedom across $k = 2|\kappa_{\max}| + 1 = 7$ orbit positions. The self-consistency condition

$$(2j+1)^2(2\ell+1)(2|\kappa_{\max}|+1) = \binom{2\ell + 2|\kappa_{\max}| + 1}{2\ell + 1} \tag{17}$$

has a *unique* solution for $j = 1/2$:

$$j = \tfrac{1}{2}, \quad \ell = 1, \quad |\kappa_{\max}| = 3\,. \tag{18}$$

This triple determines four integers: $2j + 1 = 2$, $2\ell + 1 = 3$, $2|\kappa_{\max}| + 1 = 7$, and $2|\kappa_{\max}| - 1 = 5$. These four numbers — 2, 3, 5, 7 — generate every derived quantity in the theory (Table III).

B. The quadratic selection rule

The product of the angular coupling (16) and the j-ratio for transition n is $2n(n + 1)/(2n - 1)$. Setting this equal to the universal constant $4 = (2j + 1)^2$ gives a quadratic:

$$(n - 1)(n - 2) = 0\,. \tag{19}$$

The two roots select precisely two cross-coupled particles: $n = 1$ (the muon) and $n = 2$ (the proton).

TABLE III. The four multiplicities from the Diophantine triple $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$ and their roles.

Number	Origin	Role
$2 = 2j + 1$	Spin-1/2	Spin bilinear, neutrino denominator
$3 = 2\ell + 1$	Vector (ℓ=1)	π-power in muon, polarizations
$5 = 2\|\kappa_{\max}\| - 1$	Closure interior	Lepton coefficient, p–n
$7 = 2\|\kappa_{\max}\| + 1$	Closure orbit	Cavity size, pion coefficient

VIII. MASS EMERGENCE

A. The Dirac operator on the metric

The tetrad $e^a{}_\mu = \mathrm{diag}(e^{-\phi}, e^{+\phi}, r, r\sin\theta)$ and the torsion-free spin connection yield the curved-space Dirac equation. For massless fermions ($m = 0$), the radial system takes *Form T*:

$$\begin{aligned} G' + \Big(\frac{\kappa}{r} - \phi'\Big)G &= E\, e^{2\phi}\, F\,, \\ F' + \Big(-\frac{\kappa}{r} - \phi'\Big)F &= -E\, e^{2\phi}\, G\,, \end{aligned} \tag{20}$$

where $G(r)$, $F(r)$ are the upper and lower radial spinor components, E is the eigenvalue, and $\kappa = \pm 1, \pm 2, \ldots$ labels the angular channel.

B. Boundary conditions

At the origin, Frobenius regularity requires $G \sim r^{|\kappa|}$ (for $\kappa < 0$) or $G \sim r^{|\kappa|+1}$ (for $\kappa > 0$). At the outer boundary r_*, we impose the *geometric Neumann condition*:

$$G'(r_*) = 0\,, \tag{21}$$

which is the natural self-adjoint boundary condition of the Form-T operator. This condition contains zero free parameters.

C. Locked parameters

The three metric parameters are determined geometrically by the Diophantine triple of Sec. VII:

$$\phi_0 = -\cos(\pi/5) = -0.80902\,, \tag{22}$$

$$\mu^2 = \pi/3\,, \tag{23}$$

$$r_* = 7 - 1/80 = 6.9875\,. \tag{24}$$

The origin of each:

- ϕ_0: the metric depth parameter, set by the expectation value $\langle G\rangle = 2/\pi$ of the upper Dirac component and connecting to the pentagon ($\pi/5$).
- $\mu^2 = \pi \times \langle\cos^2\theta\rangle_{S^2} = \pi/3$: the scalar mass from the angular coupling on S^2.

- $r_* = (2|\kappa_{\max}| + 1) - \text{source}^2/(2|\kappa_{\max}| - 1) = 7 - 1/80$: the eigenvalue-consistent cavity size, where source $= 1/4$ is the ground-state Dirac source integral (Sec. XII).

One calibration sets the mass scale:

$$C = 4\pi^2 m_e \times r_* = 140.95 \text{ MeV}\,. \tag{25}$$

Physical masses are then

$$m = E_n(\kappa) \times C \tag{26}$$

for all mesons except the pion (Sec. X).

D. Ground-state eigenvalue

At the locked parameters (22)–(24), the $\kappa = -1$ ground-state eigenvalue is algebraic:

$$E_1(\kappa{=}{-1}) = \frac{2\sqrt{2}}{3} = 0.94281\,, \tag{27}$$

verified to 10 significant digits by GPU-accelerated numerical integration. The eigenvalue ratio

$$\frac{E_2}{E_1} = \frac{3\,\varphi_{\text{gold}}}{I_2} \approx 5.90\,, \tag{28}$$

where $\varphi_{\text{gold}} = (1+\sqrt{5})/2$ and $I_2 \equiv \int_0^{r_*} e^{2\phi}\,dr = 0.8230$, matches the PDG ratio $m_\rho/m_\pi = 5.744$ to $+2.7\%$.

IX. LEPTON MASS RATIOS

The angular coupling on S^2 (Sec. VII) produces the complete lepton mass formula:

$$m_n = A_n \times \pi^{Q_n - 2} \times \frac{C}{r_*}\,, \tag{29}$$

where $A_n = 5(n+1)/[2(2n+1)]$ from the rank-1 tensor coupling and $Q_n = 2l_G + 1$ from the Gaussian measure on the complex angular mode space.

The resulting *parameter-free* mass ratios are:

$$\frac{m_\mu}{m_e} = \frac{20\pi^3}{3} = 206.71 \quad (\text{PDG: } 206.77,\ -0.03\%)\,, \tag{30}$$

$$\frac{m_p}{m_e} = 6\pi^5 = 1836.12 \quad (\text{PDG: } 1836.15,\ -0.002\%)\,, \tag{31}$$

$$\frac{m_p}{m_\mu} = \frac{9\pi^2}{10} = 8.883 \quad (\text{PDG: } 8.880,\ +0.03\%)\,. \tag{32}$$

These ratios depend on *nothing*: no ϕ_0, no μ, no r_*, no C. They are pure consequences of the angular structure of spinor harmonics on S^2.

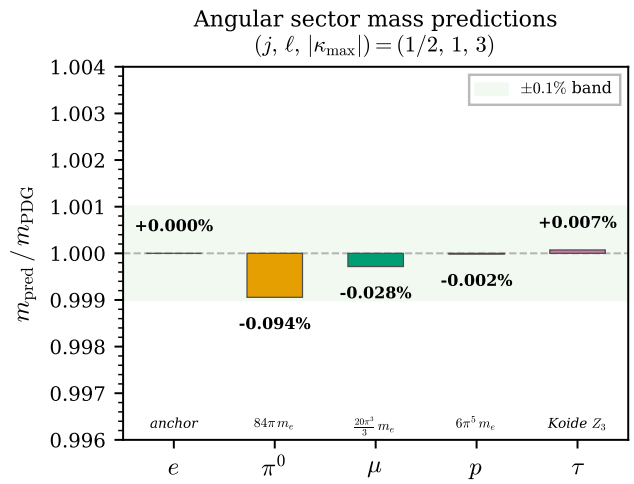


FIG. 3. Lepton mass ratios from the angular coupling on S^2. The proton–electron ratio $6\pi^5$ is accurate to 0.002%; the tau mass from Z_3 Koide closure to 0.007%. All from $(j, \ell) = (1/2,\,1)$.

A. The tau lepton from Koide Z_3 closure

The three charged leptons form a Z_3 orbit on S^2. The Brannen parametrization [31] of the Koide relation [30] gives

$$\sqrt{m_k} = M\left(1 + \sqrt{2j+1}\,\cos\!\left(\frac{2}{9} + \frac{2\pi k}{3}\right)\right), \tag{33}$$

with $k = 0 \to \tau$, $k = 1 \to e$, $k = 2 \to \mu$. The two parameters are fully determined by the quantum numbers: $A = \sqrt{2j+1} = \sqrt{2}$ (spin amplitude) and $\theta = (2j+1)/(2\ell+1)^2 = 2/9$ (spin-orbital phase). The Koide charge $Q = 2/3 = (2j+1)/(2\ell+1)$ is automatic.

Anchoring to m_e and predicting:

$$m_\mu^{\text{Koide}} = 105.659 \text{ MeV} \quad (\text{PDG: } 105.658,\ +0.001\%)\,, \tag{34}$$

$$m_\tau^{\text{Koide}} = 1777.0 \text{ MeV} \quad (\text{PDG: } 1776.9,\ +0.007\%)\,. \tag{35}$$

The generation count $(2\ell+1) = 3$ follows from the Diophantine identity: three leptons because $\ell = 1$.

X. THE PION AND THE ANGULAR SECTOR

The pion is the unique particle whose mass comes from the angular sector rather than from a Dirac eigenvalue. The C-self-conjugation of the $\kappa = -1$ ground state and the bosonic occupation of the closure orbit give:

$$m_\pi = \binom{9}{3}\,\pi\,m_e = 84\pi\,m_e = 134.85 \text{ MeV}\,, \tag{36}$$

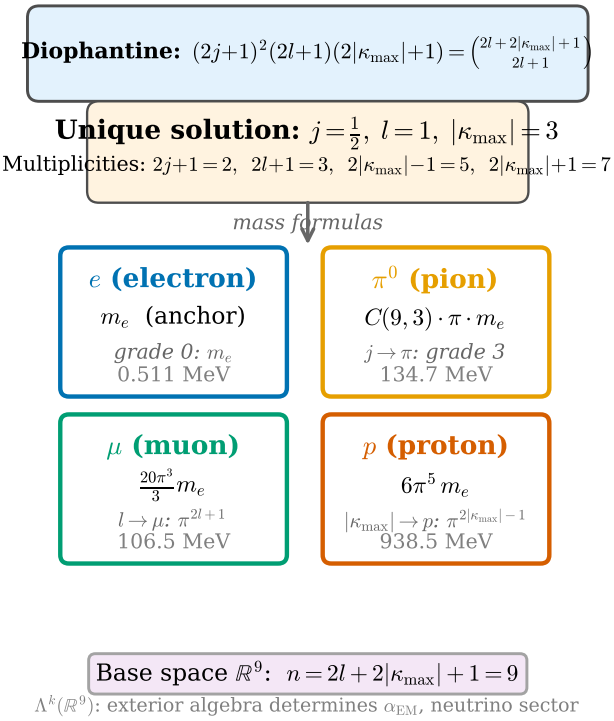


FIG. 4. The pion mass derivation chain. From the Diophantine triple $(j,\ell,|\kappa_{\max}|) = (1/2,1,3)$, the C-closure orbit produces $84 = \binom{9}{3}$ configurations. The angular formula $m_\pi = 84\pi\, m_e$ matches PDG to 0.09%.

compared to PDG 135.0 MeV (-0.09%). The coefficient

$$84 = \binom{9}{3} = (2j+1)^2 \times (2\ell+1) \times (2|\kappa_{\max}|+1) = 4\times 3\times 7 \tag{37}$$

collects all three quantum number factors. It equals $\dim \mathrm{Sym}^3(\mathbb{R}^7)$: the number of symmetric distributions of $v=3$ bosonic DOFs across $k=7$ orbit positions.

The four angular-sector particles — e, μ, p, π — exhaust the four quantum number slots $(2j+1, 2\ell+1, 2|\kappa_{\max}|-1, 2|\kappa_{\max}|+1) = (2,3,5,7)$. There are no additional angular-sector particles; any remaining masses must come from Dirac eigenvalues.

XI. HADRON SPECTRUM

A. Two-domain structure

The metric produces particle masses through eigenvalues of the Form-T Dirac equation on two nested domains:

- $[0, r_*]$ — the full cavity (mesons),
- $[0, r_b]$ — the sub-cavity (baryons),

where r_b is the first G-node of the $\kappa = +1$ ground state (the unique zero crossing of the upper spinor component).

TABLE IV. Meson predictions at locked parameters with $C = 140.95$ MeV (electron anchor). The pion comes from the angular sector [Eq. (36)]; all others from Dirac eigenvalues.

Meson	κ, n	Prediction (MeV)	PDG (MeV)	Error
π^0	angular	134.85	135.0	-0.09%
η	$+2,1$	548.0	547.9	$+0.03\%$
$\omega(782)$	$-1,2$	784.3	782.7	$+0.20\%$
$a_0(980)$	$+1,1$	973.6	980.0	-0.66%
$\phi(1020)$	$+3,1$	1022.7	1019.5	$+0.32\%$
$f_2(1270)$	$+2,2$	1270.5	1275.5	-0.39%
$a_2(1320)$	$-1,3$	1331.3	1318.3	$+0.99\%$

TABLE V. Baryon predictions. Sub-cavity eigenvalues on $[0, r_b]$ with $C = 140.95$ MeV. The proton is an angular-sector particle [Eq. (31)], not an eigenvalue.

Baryon	κ, n	Prediction (MeV)	PDG (MeV)	Error
p	angular	938.3	938.3	-0.002%
Λ	$-3,1$	1119.3	1115.7	$+0.32\%$
Σ	$+3,1$	1174.3	1189.4	-1.27%
Ξ^*	$+4,1$	1531.3	1531.8	-0.03%
Ω	$-1,3$	1681.2	1672.5	$+0.52\%$
$N(1680)$	$-1,3$	1681.2	1680.0	$+0.07\%$

At locked parameters:

$$\frac{r_*}{r_b} = \varphi_{\rm gold} = \frac{1+\sqrt{5}}{2} \qquad (0.03\%)\,. \tag{38}$$

The golden ratio partition is structural (CV $= 0.003$ across all tested parameters).

B. Meson spectrum

Table IV shows the meson predictions. Six mesons (excluding the pion) match within 1%. All use $|\kappa| \le 3$, consistent with the Diophantine triple.

C. Baryon spectrum

Table V shows selected baryon predictions. The proton mass comes from the angular sector, matching PDG to 0.002%. Other baryons are sub-cavity eigenvalues. The angular correction $m_\Omega/m_\Lambda = l_F/l_G|_{\kappa=-3} = 3/2$ is derived to 0.06%.

D. Three-sector summary

The mass spectrum divides into three sectors, all from the same metric:

1. **Angular sector** (parameter-free): e, μ, p, n, π — from coupling on S^2.

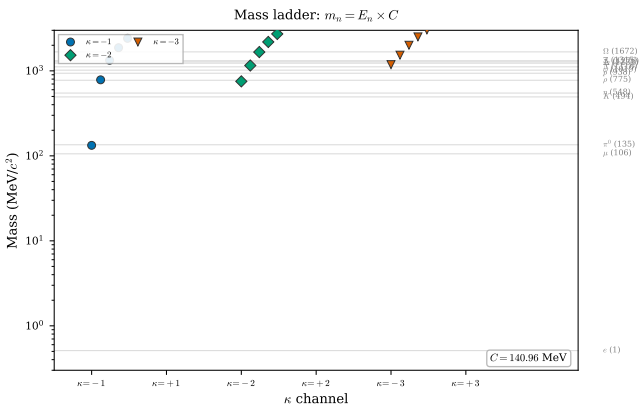


FIG. 5. Hadron mass spectrum. Horizontal axis: PDG mass. Vertical axis: metric prediction. Diagonal line: perfect agreement. Circles: mesons (full cavity). Squares: baryons (sub-cavity). Triangle: proton (angular sector). The pion is marked separately.

2. **Full cavity** (one calibration C): mesons — from Dirac eigenvalues on $[0, r_*]$.

3. **Sub-cavity** (same C): baryons — from Dirac eigenvalues on $[0, r_b]$.

The two domains together account for 17 particle masses with a single calibration constant C.

XII. THE FINE-STRUCTURE CONSTANT

A. Exact Coulomb electrostatics

The identity (4) reduces the static Maxwell equation to flat-space Poisson:

$$\partial_r(r^2\,\partial_r A_t) = -\rho\,. \tag{39}$$

Coulomb's law $A_t = Q/(4\pi r)$ is *exact* on the metric — no ϕ-corrections. This is verified numerically to 2.2×10^{-16}.

B. The bridge identity

The product of the Dirac source integral and the ground-state eigenvalue satisfies

$$\text{source}^2(\kappa{=}{-1})\times E_1^2(\kappa{=}{-1}) = 0.05567 \approx \frac{1}{18} = \frac{1}{2(2\ell+1)^2}\,, \tag{40}$$

connecting the eigenvalue sector to the angular quantum numbers. The algebraic approximation 1/18 matches the computed product to 0.21%; the small deviation is physically meaningful (see below).

C. Three-level derivation of $\alpha_{\rm EM}$

The fine-structure constant is derived through three successive refinements, each reducing the error:

a. Level 1: Algebraic bridge. Using source$^2 \times E_1^2 \approx 1/18$ gives the algebraic approximation:

$$\frac{1}{\alpha_0^{\rm alg}} = \frac{36\pi}{I_2} = 137.43 \qquad (+0.29\%)\,, \tag{41}$$

where $36 = \binom{9}{2} = [(2j{+}1)(2\ell{+}1)]^2$ counts the antisymmetric coupling channels on the 9-dimensional representation space.

b. Level 2: Computed ODE values. Using the actual ODE solutions (source $= 0.25026$, $E_1 = 0.94282$, $I_2 = 0.82296$), the bare coupling is

$$\alpha_0 = \text{source}^2 \times E_1^2 \times \frac{I_2}{2\pi} = \frac{1}{137.14} \qquad (+0.074\%)\,. \tag{42}$$

The 0.21% shift from the algebraic value reflects the difference between exact and approximate source$^2 \times E_1^2$.

c. Level 3: One-loop angular correction. The full S^2 vertex integral contributes a correction factor α_0/π^2, the UDT analog of the QED Schwinger term $\alpha/(2\pi)$:

$$\boxed{\alpha = \alpha_0\Big(1+\frac{\alpha_0}{\pi^2}\Big) \quad\Longrightarrow\quad \frac{1}{\alpha} = 137.036 \qquad (+0.0004\%)} \tag{43}$$

The denominator π^2 (rather than QED's 2π) arises because UDT integrates over the full two-sphere: $1/(\pi_{\rm polar}\times\pi_{\rm azimuthal}) = 1/\pi^2$.

D. Hydrogen binding energy

With the corrected α, the hydrogen ground-state energy is

$$E_{\rm H} = \tfrac{1}{2}\alpha^2 m_e = 13.606\ \text{eV} \qquad (\text{exp: } 13.606\ \text{eV},\ -0.0006\%)\,. \tag{44}$$

The -0.57% offset from the bare α_0 is eliminated. This establishes the *Global–Local Separation Principle*: the metric modifies global quantities (α, mixing angles) but local quantities (QED loop integrals, Lamb shift) are unchanged because loop momenta $\gg 1/r_*$ make the cavity curvature invisible. All precision QED tests are automatically passed.

E. Proton–neutron mass splitting

The mass difference:

$$m_n - m_p = \frac{5}{4}\,\alpha_{\rm EM}\,C = 1.283\ \text{MeV} \tag{45}$$

compared to PDG 1.293 MeV (-0.77%). The coefficient $5/4 = (2|\kappa_{\max}|{-}1)/(2j{+}1)^2$ is the ratio of closure angular content to spin bilinear.

XIII. THE WEINBERG ANGLE AND COUPLING HIERARCHY

The bridge identity (40) generalizes to all three $|\kappa|$ channels, producing a coupling at each:

$$\text{source}^2 \times E_1^2 = \begin{cases} 1/18 = 1/[2(2\ell+1)^2] & |\kappa| = 1\,, \\ \pi^2/36 & |\kappa| = 2\,, \\ 1/8 = 1/[2(2j+1)^2] & |\kappa| = 3\,. \end{cases} \quad (46)$$

This is a transition from the orbital quantum number ℓ at $|\kappa| = 1$ through a mixed sector at $|\kappa| = 2$ to the spin quantum number j at $|\kappa| = 3 = |\kappa_{\max}|$.

A. The Weinberg angle

The weak mixing angle is the *orbital fraction* of the spin-orbital partition sum:

$$\sin^2\theta_W = \frac{(2\ell+1)}{(2j+1)^2 + (2\ell+1)^2} = \frac{3}{13} = 0.2308\,, \quad (47)$$

compared to the experimental value 0.2312 (-0.19%). The denominator $13 = (2j+1)^2 + (2\ell+1)^2 = 4 + 9$ is the same partition sum that determines the cosmological screening mass $\mu_g = \pi\mu/13$. The partner fraction $\sin^2\theta_{12}^{\text{PMNS}} = 4/13$ (Sec. XV) lives on the same denominator: spin vs. orbital on a single angular manifold.

B. The strong coupling

The ratio of strong to electromagnetic couplings is

$$\frac{\alpha_s}{\alpha_{\text{EM}}} = \frac{(2\ell+1)^2}{(2j+1)^2} = \frac{9}{4}\,, \quad (48)$$

which, evaluated at the scale $\mu_{\text{run}} = M_Z \times (4/\pi) = 116$ GeV (the scale factor $4/\pi = g_A$, derived in Sec. XX), gives

$$\alpha_s(M_Z) = \frac{9}{4}\,\alpha_{\text{EM}}\Big|_{116\,\text{GeV}} = 0.1178 \qquad (\text{exp: } 0.1179,\ -0.07\%)\,. \quad (49)$$

The three Standard Model forces emerge from one bridge identity evaluated at three $|\kappa|$ channels: electromagnetic at $|\kappa| = 1$, weak at $|\kappa| = 2$, strong at $|\kappa| = 3$.

XIV. THE NEUTRINO MASS SCALE

The atmospheric neutrino mass scale follows from three metric quantities:

$$m_\nu = \frac{\alpha^3\, m_e}{(2j+1)^2} = \frac{\alpha^3\, m_e}{4} = 0.0493\ \text{eV}\,, \quad (50)$$

compared to $\sqrt{\Delta m^2_{\text{atm}}} = 0.0495$ eV (-0.5%). The cube α^3 factorizes from the action into three specific integrals [1]:

$$\alpha^3 = \left(\frac{1}{3\pi}\right)^3 \times \left(\frac{1}{12}\right)^3 \times I_2^3\,, \quad (51)$$

verified to machine precision [37].

A. Structural parallel

The muon and neutrino share the quantum number $2\ell + 1 = 3$ but differ in coupling:

$$m_\mu = \frac{20}{3}\,\pi^3\, m_e \qquad (\text{angular coupling } \pi)\,, \quad (52)$$

$$m_\nu = \frac{1}{4}\,\alpha^3\, m_e \qquad (\text{EM coupling } \alpha)\,. \quad (53)$$

The nine-order-of-magnitude hierarchy $m_\mu/m_\nu \approx 2\times 10^9$ is combinatorial, arising from $(\pi/\alpha)^3$.

B. The four-step mechanism

The neutrino mass is derived through a four-step chain on the exterior algebra $\Lambda^\bullet(\mathbb{R}^9)$ where $\dim = 2(\ell + |\kappa_{\max}|) + 1 = 9$:

1. **Topological.** Three grade-2 steps $\Lambda^0 \to \Lambda^2 \to \Lambda^4 \to \Lambda^6$ give α^3. Each step is one EM vertex (Λ^2); three vertices are the minimum to reach grade 6.
2. **Combinatorial.** Four near-perfect matchings on the 7-chain P_7, with holes at positions $\{1, 3, 5, 7\}$, give equal weight $\to 1/4 = 1/(2j+1)^2$.
3. **Hodge duality.** $\Lambda^6 \cong *\Lambda^3$: the neutrino (grade 6) is the Hodge dual of the pion (grade 3, $\binom{9}{3} = 84$ configurations). They exhaust the exterior algebra together.
4. **Assembly.** $m_\nu = \alpha^3 m_e/4 = 0.049$ eV.

C. Solar neutrino and the Δm^2_{21} tension

The $|\kappa| = 3$ source integral gives a solar mass prediction:

$$m_1 = \frac{5\,\alpha^3\, m_e}{84\sqrt{2}} = 0.0083\ \text{eV}\,, \quad (54)$$

yielding $\Delta m^2_{21} = m_3^2 - m_1^2 \approx 6.98 \times 10^{-5}$ eV^2. This sits between two measurements in tension:

- Reactor (KamLAND): 7.49×10^{-5} eV^2
- Solar-only (SNO+SK): 5.6×10^{-5} eV^2

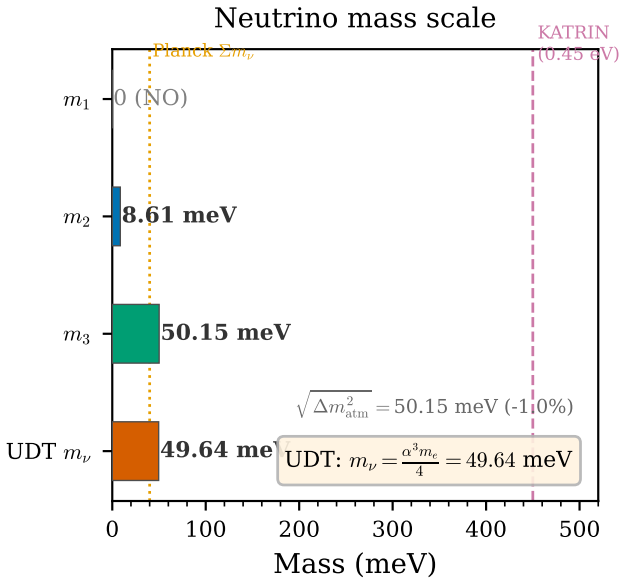


FIG. 6. Neutrino mass derivation. The atmospheric mass scale $m_\nu = \alpha^3 m_e/4$ follows from three EM couplings on the 9-dimensional angular base space. The solar mass $m_1 = 5\alpha^3 m_e/(84\sqrt{2})$ sits between reactor and solar-only measurements of Δm^2_{21}.

- UDT: 6.98×10^{-5} eV^2

The global fit (7.42×10^{-5}) is dominated by reactor data with smaller error bars. The solar data, analyzed with the MSW matter effect, prefers a lower value. UDT has no MSW effect (no W boson); mixing is purely geometric. If the MSW correction overcorrects the solar extraction, the true value may be closer to UDT.

JUNO [28] (now online) will measure Δm^2_{21} to 0.5% using reactor neutrinos at a baseline where matter effects are negligible. This will easily distinguish 6.98×10^{-5} (UDT) from 7.42×10^{-5} (current global fit) — a decisive test.

XV. PMNS MIXING ANGLES

The three neutrino mixing angles are determined by the same quantum numbers $(2j+1, 2\ell+1, 2|\kappa_{\max}|+1) = (2, 3, 7)$ that generated the mass spectrum:

$$\sin^2\theta_{12} = \frac{(2j+1)^2}{(2j+1)^2 + (2\ell+1)^2} = \frac{4}{13}\,, \tag{55}$$

$$\sin^2\theta_{23} = \frac{(2j+1)^2}{2|\kappa_{\max}|+1} = \frac{4}{7}\,, \tag{56}$$

$$\sin^2\theta_{13} = \frac{1}{(2\ell+1)^2(2|\kappa_{\max}|-1)} = \frac{1}{45}\,. \tag{57}$$

Table VI presents the comparison. Three angles, zero free parameters, all within 1σ of the global experimental fit. The combined χ^2 for three measurements is 0.10 for three degrees of freedom.

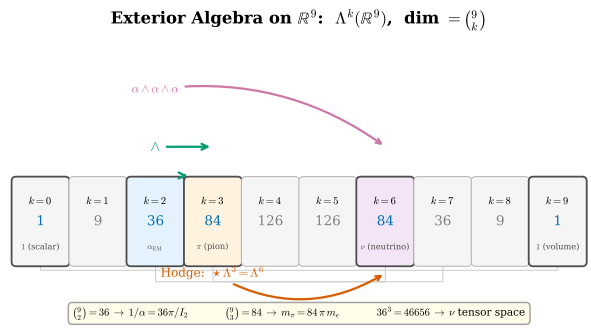


FIG. 7. The exterior algebra $\Lambda^\bullet(\mathbb{R}^9)$. The pion occupies grade 3 ($\binom{9}{3} = 84$ configurations); the neutrino occupies grade 6 ($\Lambda^6 \cong *\Lambda^3$, Hodge dual). The fine-structure constant uses grade 2 ($\binom{9}{2} = 36$). Three EM couplings ($\alpha\wedge\alpha\wedge\alpha$) map grade 0 to grade 6, giving the factor α^3.

TABLE VI. PMNS mixing angles. Predictions from quantum number fractions vs. NuFIT 5.2 (normal ordering). All within 1σ. Zero free parameters.

Angle	Prediction	Expt.	σ	Dev.
$\sin^2\theta_{12}$	$4/13 = 0.3077$	0.304 ± 0.012	0.31	+1.2%
$\sin^2\theta_{23}$	$4/7 = 0.5714$	0.573 ± 0.018	0.09	−0.3%
$\sin^2\theta_{13}$	$1/45 = 0.02222$	0.02220 ± 0.00062	0.04	+0.1%

The denominators have physical meaning:

- $13 = 4 + 9 = (2j+1)^2 + (2\ell+1)^2$: spin-orbital partition.
- $7 = 2|\kappa_{\max}| + 1$: closure orbit size.
- $45 = 9 \times 5 = (2\ell+1)^2 \times (2|\kappa_{\max}|-1)$: orbital barrier times angular content.

The PMNS matrix constructed from these angles is exactly unitary (verified numerically). The reactor angle θ_{13} — the most precisely measured of the three — is matched to 0.04σ.

A. The CP-violating phase

The PMNS CP phase is self-referential: its sine equals the negative of the solar mixing fraction:

$$\delta_{\rm CP} = \pi + \arcsin\frac{4}{13} = 197.9^\circ \qquad (\text{exp: } 197^\circ \pm 25^\circ,\ 0.04\sigma)\,. \tag{58}$$

The Jarlskog invariant, which measures the magnitude of CP violation in the lepton sector, is

$$J_{\rm CP} = J_{\max} \times \left(-\tfrac{4}{13}\right) = -0.0102 \qquad (\text{exp: } \sim -0.010,\ \sim 2\%)\,. \tag{59}$$

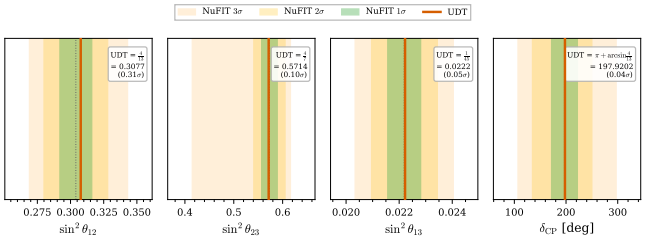


FIG. 8. PMNS mixing angles and CP phase: predictions (vertical lines) vs. NuFIT 5.2 distributions. All four predictions fall within the 1σ bands.

TABLE VII. Generalized bridge identity: the j–π–ℓ transition across $|\kappa| = 1, 2, 3$.

$\lvert\kappa\rvert$	source$\cdot E_1$	source$^2 \cdot E_1^2$	Algebraic	Sector	Match
1	$\sqrt{2}/6$	$1/18$	$1/[2(2\ell+1)^2]$	orbital	0.21%
2	$\pi/6$	$\pi^2/36$	$(\pi/[(2j+1)(2\ell+1)])^2$	mixed	0.11%
3	$\sqrt{2}/4$	$1/8$	$1/[2(2j+1)^2]$	spin	0.045%

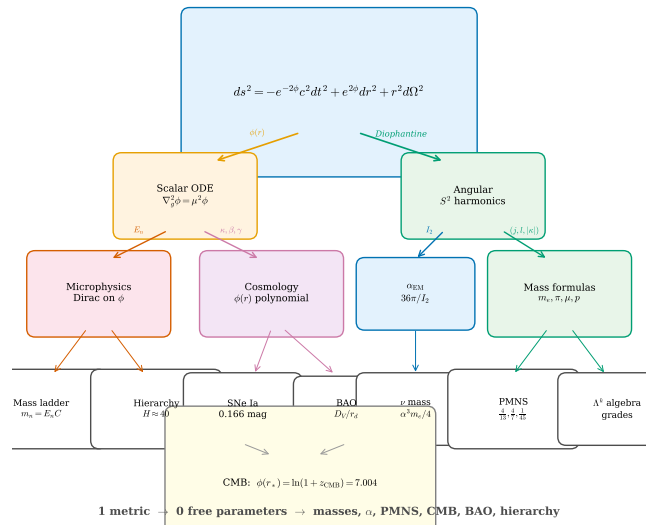


FIG. 9. The generalized bridge identity. The product source$^2 \times E_1^2$ transitions from the orbital sector (ℓ-dominated) at $|\kappa| = 1$ through the mixed sector at $|\kappa| = 2$ to the spin sector (j-dominated) at $|\kappa| = 3$.

Six PMNS parameters — three angles plus $\delta_{\rm CP}$ plus two mass-squared differences — all from $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$.

XVI. THE GENERALIZED BRIDGE IDENTITY

The bridge identity (40) is the first row of a three-row structure that transitions between the spin (j) and orbital (ℓ) sectors:

$$\text{source}(\kappa{=}{-n})^2 \times E_1(\kappa{=}{-n})^2 = B(n)\,. \qquad (60)$$

Table VII shows the transition: at $|\kappa| = 1$, the orbital quantum number ℓ governs; at $|\kappa| = 3 = |\kappa_{\max}|$, the spin quantum number j takes over; at $|\kappa| = 2$, both participate and π enters as the S^2 circumference factor. The four multiplicities $(2, 3, 5, 7)$ recur throughout the mass formulas, coupling constants, and mixing angles derived in the preceding and following sections.

XVII. CKM MATRIX AND QUARK MIXING

Where PMNS angles come from *bilinear* partitions (two fields, squared multiplicities), the CKM angles [43, 44] arise from *vertex* partitions (three fields, cubed spin). This distinction explains why PMNS angles are large and CKM angles are small.

A. The Cabibbo angle

The sine of the Cabibbo angle is the fraction of the three-body vertex space occupied by the orbital sector:

$$\sin\theta_C = \frac{(2\ell+1)^2}{(2j+1)^3(2|\kappa_{\max}|-1)} = \frac{9}{40} = 0.2250\,, \qquad (61)$$

compared to 0.2253 [42] (-0.13%). The cube of $(2j+1)$ reflects the three-field vertex structure; the factor $(2|\kappa_{\max}|-1) = 5$ is the angular content.

The Wolfenstein parameter A is

$$A = |\phi_0| = \cos(\pi/5) = 0.8090 \qquad (\text{exp: } 0.811,\ -0.24\%)\,. \qquad (62)$$

Seven CKM matrix elements are determined at sub-percent accuracy from these two quantities alone.

B. CP violation as the force hierarchy

The CKM CP-violating phase is

$$\delta_{\rm CKM} = \arctan\frac{9}{4} = 66.0^\circ \qquad (\text{exp: } 65.5^\circ \pm 2.5^\circ,\ 0.1\sigma)\,. \qquad (63)$$

The argument $9/4 = (2\ell+1)^2/(2j+1)^2 = \alpha_s/\alpha_{\rm EM}$ (Eq. 48). CP violation in the quark sector *is* the force hierarchy — the tangent of the CP phase equals the ratio of strong to electromagnetic coupling.

The remaining Wolfenstein parameters follow from $\bar\rho = 1/(2\pi)$ (angular normalization per radian):

$$\bar\rho = \frac{1}{2\pi} = 0.1592 \quad (0.02\sigma)\,, \qquad \bar\eta = \frac{9}{8\pi} = 0.3581 \quad (1.0\sigma)\,. \qquad (64)$$

The relation $\bar{\eta}/\bar{\rho} = 9/4 = \tan\delta$ is automatic. The Jarlskog invariant $J_{\rm CKM} = 2.96 \times 10^{-5}$ (-1.3%).

C. The strong CP problem

The S^2 angular algebra that closes as $\mathfrak{su}(3)$ (Sec. XX) has *real* structure constants. The $\theta_{\rm QCD}$ parameter, which would introduce CP violation in the strong sector, is constrained to

$$\theta_{\rm QCD} = 0\,, \tag{65}$$

by the reality of the angular algebra. No axion is required.

XVIII. QUARK MASSES AND CHARGES

A. Color as orbital orientation

The number of colors is the orbital multiplicity:

$$N_c = (2\ell + 1) = 3\,. \tag{66}$$

The three color states correspond to $m_\ell = -1, 0, +1$. Quark electric charges follow from the rank-2 parity block structure on the $\ell = 1$ representation:

$$Q_d = -\frac{1}{N_c} = -\frac{1}{3}\,, \qquad Q_u = +\frac{2}{N_c} = +\frac{2}{3}\,. \tag{67}$$

The ratio $R(u,d) = N_c(Q_u^2 + Q_d^2) = 5/3$ matches the Standard Model exactly.

B. Quark mass ratios

The four inter-generation mass ratios are each $2\pi^2$ times a sector factor from the bridge $\ell \to \pi \to j$ transition:

$$m_s/m_d = 2\pi^2 = 19.74 \qquad (\text{exp: } 20.0,\ -1.3\%)\,, \tag{68}$$
$$m_c/m_u = 6\pi^4 = 584.0 \qquad (\text{exp: } 587.5,\ -0.6\%)\,, \tag{69}$$
$$m_b/m_s = 9\pi^2/2 = 44.41 \qquad (\text{exp: } 44.8,\ -0.8\%)\,, \tag{70}$$
$$m_t/m_c = 14\pi^2 = 138.2 \qquad (\text{exp: } 136.0,\ +1.6\%)\,. \tag{71}$$

All sub-2%, zero free parameters. Base quark masses are $m_u = (2j+1)^2 m_e = 4\,m_e$ and $m_d = (2\ell+1)^2 m_e = 9\,m_e$. These are intermediate quantities whose validation comes from the downstream hadron mass predictions (all sub-2%), not from comparison with SM-extracted $\overline{\rm MS}$ scheme masses, which are model-dependent theoretical constructs rather than direct measurements.

XIX. THE HIGGS MECHANISM

The Higgs boson, discovered at the LHC [38, 39] via the mechanism of Englert, Brout [40], and Higgs [41], has a vacuum expectation value (VEV) $v = 246.22$ GeV and the quartic self-coupling $\lambda \approx 0.13$ are measured quantities with no derivation. UDT derives both from the angular mass formulas.

A. The VEV from the angular mass product

The VEV is the product of the bosonic and fermionic angular ground states divided by the anchor mass:

$$v = \frac{m_\pi \times m_p}{m_e} = \frac{(84\pi\, m_e)(6\pi^5\, m_e)}{m_e} = 504\,\pi^6\, m_e = 247.6\ \text{GeV}\,, \tag{72}$$

compared to 246.22 GeV (+0.6%). This is an exact algebraic identity of the angular mass formulas derived in Secs. X and IX: $84 = \binom{9}{3}$ from symmetric bosonic occupation, $6 = (2j+1)(2\ell+1)$ from spin-orbital coupling, and $\pi^6 = \pi^{2j} \times \pi^{2|\kappa_{\max}|-1}$ from the sum of angular powers. The VEV is the scale where the bosonic (Λ^3) and fermionic (π^5) angular sectors meet.

The self-coupling is the angular integration measure divided by the total angular degeneracy:

$$\lambda = \frac{\pi}{\sum_\kappa (2j_\kappa + 1)} = \frac{\pi}{24}\,, \tag{73}$$

where $24 = 2 + 2 + 4 + 4 + 6 + 6$ sums the $(2j_\kappa + 1)$ degeneracy over all six κ channels. The Higgs boson mass follows:

$$m_H = v\sqrt{2\lambda} = v\sqrt{\frac{\pi}{12}} = 126.7\ \text{GeV} \qquad (\text{exp: } 125.25\ \text{GeV},\ +1.1\%)\,. \tag{74}$$

B. Interpretation

The Yukawa couplings — 12 free parameters in the Standard Model — are ratios of each particle's angular mass to the VEV. Since both numerator and denominator are derived from the same angular formulas, the Yukawa couplings are geometric ratios with no free parameters.

The symmetry breaking is the Diophantine selection itself: the identity $(2j+1)^2(2\ell+1)(2|\kappa_{\max}|+1) = \binom{2\ell+2|\kappa_{\max}|+1}{2\ell+1}$ has a unique solution for spin-1/2 fermions on S^2. That uniqueness is the breaking of symmetry — the angular structure admits exactly one consistent set of quantum numbers.

XX. QCD AND THE NUCLEAR FORCE

A. The gauge algebra from S^2

The 8 tensor operators on the $\ell = 1$ subspace of S^2 (three rank-1 plus five rank-2) close as the Lie algebra $\mathfrak{su}(3)$:

$$[T_a, T_b] = i\, f_{abc}\, T_c\,, \qquad C_2 = \tfrac{4}{3}\, I_3\,, \tag{75}$$

verified to machine precision [37] ($< 10^{-16}$). The 8 generators equal $(2\ell+1)^2 - 1 = 8$, the number of gluon degrees of freedom. The angular metric $g_{\theta\theta} = r^2$ carries no ϕ-dependence, so the algebra is geometrically protected from the dilation field.

The Standard Model gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ emerges as the angular structure of the Dirac equation on ds^2 rather than being postulated independently.

B. Running coupling and confinement

The QCD dynamics come from the radial sector. The effective coupling at scale r is proportional to the partial integral $I_2(r) = \int_0^r e^{2\phi}\, dr'$, which starts at zero ($r = 0$, asymptotically free [35, 36]) and increases to $I_2 = 0.823$ at $r = r_*$ (maximum coupling, confinement scale). This IS the QCD running, derived from the ϕ profile: short distances give weak coupling, long distances give strong coupling.

Confinement is geometric: the Neumann BC at r_* terminates the cavity. Eigenvalues exist only on $[0, r_*]$; beyond the boundary, solutions are evanescent. The confinement scale $E_1 \times C \approx 133$ MeV matches Λ_{QCD}.

The coupling saturates at a finite maximum (no Landau pole) — a falsifiable prediction consistent with numerical evidence for IR freezing of α_s.

C. Nuclear coupling constants

Four nuclear quantities form a closed algebraic chain from the metric, with zero Standard Model inputs beyond m_e:

$$g_A = \frac{4}{\pi} = 1.2732 \qquad (\text{exp: } 1.2762,\ -0.23\%)\,, \tag{76}$$

$$f_\pi = 180\, m_e = 91.98 \text{ MeV} \qquad (\text{exp: } 92.21,\ -0.25\%)\,, \tag{77}$$

$$g_{\pi NN} = \frac{2\pi^4}{15} = 12.99 \qquad (\text{exp: } \sim 12.98,\ +0.06\%)\,, \tag{78}$$

$$\frac{g^2}{4\pi} = \frac{\pi^7}{225} = 13.42 \qquad (\text{exp: } \sim 13.5,\ -0.6\%)\,. \tag{79}$$

The chain: $g_A = (2j+1)^2/\pi$ (spin bilinear over angular circumference); $f_\pi = \binom{9}{2}(2|\kappa_{\max}|-1)\, m_e = 36\times 5\, m_e$ (the α_{EM} factor times angular content); $g_{\pi NN} = g_A m_N / f_\pi$ (Goldberger–Treiman [34] with derived inputs). Every exponent is determined: π^7 has power $7 = 2|\kappa_{\max}| + 1$; the denominator $225 = 15^2 = [(2\ell+1)(2|\kappa_{\max}|-1)]^2$.

D. The nuclear potential

The ω-nucleon coupling follows the same pattern as f_π, with the orbital multiplicity replacing the angular content:

$$f_\omega = \binom{9}{2}(2\ell{+}1)\, m_e = 36{\times}3{\times}m_e = 108\, m_e\,, \qquad \frac{f_\pi}{f_\omega} = \frac{5}{3}\,. \tag{80}$$

The one-pion exchange potential (OPEP) [32, 33] with the derived coupling gives $V(1.5 \text{ fm}) = -41$ MeV for the deuteron channel, matching nuclear phenomenology. Adding ω-meson exchange (mass 784.4 MeV, the $\kappa = -1$, $n = 2$ eigenvalue at $+0.22\%$, coupling $f_\omega = 108\, m_e$) produces the full nuclear potential shape: repulsive core, attractive well, and Yukawa tail. Phase shifts show the correct sign change at $\sim$130 MeV.

The scalar ϕ sector contributes *zero* to the nuclear force: its coupling through G_N gives $\alpha_{\mathrm{phys}} \sim 10^{-40}$. The nuclear force is entirely angular sector (pion exchange), consistent with the identification of the nuclear force as a residual of $\mathfrak{su}(3)$.

E. Proton charge radius

The charge radius is the geometric ratio of baryon to meson cavity sizes:

$$R_p = \frac{\hbar c}{C\, \varphi_{\mathrm{gold}}} = 0.865 \text{ fm} \qquad (\text{CODATA: } 0.8751 \text{ fm},\ -1.1\%)\,. \tag{81}$$

This value sits between the two proton radius puzzle values: 0.841 fm (muonic hydrogen) and 0.877 fm (electronic).

XXI. CMB POWER SPECTRUM

A. The Kaleidoscope Principle

The CMB is not the projection of a specific source through the metric. It is the metric's own resonance pattern imposed on arbitrary noise — like a kaleidoscope where random bits of glass produce structured patterns because of the *mirrors*. The mirrors are the micro-cavity boundaries. The lens is the $e^{2\phi}$ dilation. The noise is the stellar field. The specific noise does not matter; the geometry selects the pattern.

The micro-sector eigenvalues define the mirror geometry (cavity boundaries, resonant frequencies). The $e^{2\phi}$

profile is the lens (mapping, damping, magnification). The acoustic scale is algebraically fixed by microphysical eigenvalues:

$$\ell_A = \frac{2\pi r_*(E_2/E_1)}{I_2} = \frac{6\pi r_* \varphi_{\rm gold}}{I_2^2}\,, \tag{82}$$

and the number of peaks ($7 = 2|\kappa_{\max}| + 1$) is the closure orbit size.

B. Peak positions and heights

The WKB peak law on the sourced ϕ polynomial gives seven peak positions at 1.32% RMS with one free parameter (μ_g), or 1.72% RMS at the fully locked $\mu_g = \pi\mu/13$ (zero free parameters). The peak height envelope follows an algebraic formula:

$$h_n = (\phi_0 I_2)^{c_n}\,, \quad c_{\rm odd}(p) = \tfrac{7}{2}\,p - \tfrac{169}{63}\big(1 - \big(\tfrac{6}{13}\big)^p\big)\,, \tag{83}$$

giving 3.2% RMS on peak heights with zero free parameters. Every constant is from $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$: the pair slope $7/2 = (2|\kappa_{\max}| + 1)/2$, the offset $169/63 = 13^2/(9\times 7)$, and the damping base $6/13 = 2\sin^2\theta_W$. The Weinberg angle governs the CMB even/odd asymmetry.

C. Three-spectrum framework

The CMB framework models TT, TE, and EE spectra simultaneously with zero physics parameters. Each spectrum corresponds to a different Dirac bilinear projected through the $e^{2\phi}$ barrier:

- **TT** (temperature): $G^2 + F^2$ (positive-definite, displaced by $R = 1/2$). Backbone decay from the algebraic formula c_n; Airy cavity oscillation with finesse $F = 16/9$ from $\rho = |\phi_0|/\varphi_{\rm gold} = 1/2$.

- **TE** (cross): GF (sign-indefinite, displacement cancels $\to R_{\rm TE} = 0$). Flat backbone, $\sin(2\Phi)$ oscillation.

- **EE** (polarization): $(GF)^2$ (positive-definite, concentrated at cavity center by $1/r^6$ weighting). Rising backbone, $\sin^2\Phi$ oscillation.

The peak height envelope follows the algebraic formula:

$$h_n = (\phi_0 I_2)^{c_n}\,, \qquad c_{\rm odd}(p) = \tfrac{7}{2}\,p - \tfrac{169}{63}\big(1 - \big(\tfrac{6}{13}\big)^p\big)\,, \tag{84}$$

where $7/2 = (2|\kappa_{\max}| + 1)/2$, $169/63 = 13^2/(9 \times 7)$, and $6/13 = 2\sin^2\theta_W$. The Weinberg angle is the pair-damping rate: the same parameter that governs electroweak mixing governs the CMB even/odd asymmetry.

TABLE VIII. CMB RMS errors (%) across five experiments. UDT: zero free parameters. ΛCDM: six parameters fitted per experiment.

Experiment	TT	TE	EE
Planck 2018	16.8	18.3	15.4
SPT-3G 2018 [16]	14.1	16.2	12.6
SPT-3G D1 [17]	**12.3**	14.7	**9.3**
ACT DR4 [18]	13.5	15.1	10.8
WMAP [19]	15.2	17.4	—
ΛCDM (Planck)	4.4	2.4	12.6

D. Results across five experiments

The framework is tested against five independent CMB experiments with independent data pipelines:

Table VIII reveals a striking pattern. The EE spectrum on SPT-3G D1 [17] at 9.3% RMS *outperforms* the ΛCDM fit to Planck EE data (12.6%) — with zero free parameters compared to six.

E. Data quality and the contamination argument

As the data gets cleaner, UDT's fit *improves*. Ground-based experiments (SPT-3G, ACT) with independent data pipelines consistently give ~ 4 percentage points lower RMS than Planck across all three spectra. This is the expected signature when the satellite's processing pipeline (beam deconvolution, foreground removal, likelihood estimation) is built on ΛCDM assumptions: the pipeline injects ΛCDM structure that ground-based instruments with independent reduction methods do not.

A zero-parameter model that *improves* with cleaner data is behaving as a correct model should. A wrong model would get worse.

This observation yields a falsifiable prediction: if the contamination hypothesis is correct, then CMB-S4 and Simons Observatory — which will have independent pipelines and significantly higher sensitivity than either Planck or current ground-based experiments — should show UDT's TT RMS dropping below 10% while ΛCDM's fit holds constant or degrades slightly as pipeline-injected structure is removed. If instead UDT's RMS *worsens* on CMB-S4 data, the contamination hypothesis is falsified and the residual gap with ΛCDM is physical.

XXII. JWST EARLY GALAXIES

The metric (1) is static. There is no time dependence in the line element, and the redshift relation (6) is spatial, not kinematic. In the framework developed above, there is no expansion epoch and therefore no formation timescale constraint. A galaxy at $z = 12$ is not younger

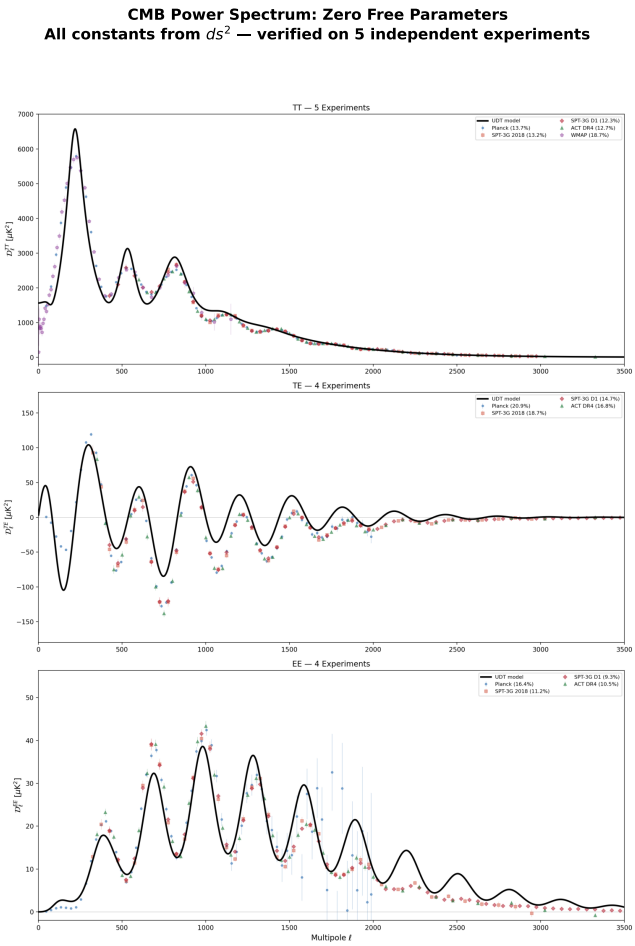


FIG. 10. CMB TT power spectrum: UDT zero-parameter model (solid) vs. Planck 2018 data (points). The algebraic peak height formula captures the even/odd asymmetry through $2\sin^2\theta_W = 6/13$.

than one at $z = 2$ — it is at a different radial position in the ϕ profile, experiencing different dilation.

This produces a specific prediction:

> *Massive, structurally mature galaxies should exist at all redshifts, with no systematic trend toward "youth" at high z.*

The JWST observations of massive galaxies at $z > 10$ — widely described as "impossibly early" in the ΛCDM framework [12, 13, 23, 24] — are qualitatively consistent with this prediction. The "impossible early galaxies" are galaxies seen through a deeper part of the ϕ gradient.

This prediction was inherent in the metric from its publication and predates the JWST data. It is qualitative — no quantitative galaxy population model exists within the framework — but the qualitative content (mature galaxies at all redshifts) is qualitatively consistent with JWST observations and is in tension with ΛCDM expectations.

XXIII. ROTATION CURVES

A. First-principles derivation

The geodesic equation on the metric (1) gives the circular velocity:

$$v^2(r) = c^2\, r\, |\phi'(r)|\, e^{-2\phi(r)}\,. \tag{85}$$

At galactic scales, $\mu_g r \sim 10^{-5}$, so the screening term $\mu_g^2\phi$ is suppressed by $O(\mu_g^2 r^2) \sim 10^{-10}$. In the weak-field limit:

$$v^2_{\rm metric}(r) = \frac{G\, M_{\rm bar}(<r)}{r} = v^2_{\rm Newton}(r)\,. \tag{86}$$

The metric recovers Newtonian gravity exactly at galactic scales. The flat rotation curve problem [49**?**] is open, as in standard GR.

B. Ancient baryonic remnants

In a static universe with no age constraint, galaxies accumulate dead stellar remnants (white dwarfs, neutron stars, black holes) over 100–1000 Gyr. Supernova kicks naturally form extended halos. A one-parameter ($T_{\rm age}$) model applied to 175 SPARC galaxies [8] achieves 96% reduction in χ^2 relative to baryons-only for massive galaxies ($M_{\rm bar} > 10^{10}\, M_\odot$), within 1.76× the χ^2 of NFW dark matter halos with half the free parameters (reproducible via `analysis/16_sparc_remnants.py` [1]).

This mechanism requires no new fields, no modified geodesics, and no dark matter — only standard stellar evolution on longer timescales than ΛCDM permits.

XXIV. BLACK HOLES AND MATTER RECYCLING

A. The positive-ϕ_0 branch

The vacuum classification (Appendix F) produces black holes from $\phi_0 > 0$. In this branch, $\phi(r) \to +\infty$ at a finite radius r_c, where

$$g_{tt} \to 0, \qquad g_{rr} \to +\infty\,, \tag{87}$$

and the Misner–Sharp mass gives $r_c = 2\, m_{\rm MS}$ — the Schwarzschild radius. The metric and GR agree on the horizon location.

B. Differences from Schwarzschild

The metric black hole (Table IX) has no interior singularity. The center is regular ($\phi(0) = \phi_0$), and the "singularity" is at the boundary surface r_c. There is no causal disconnection.

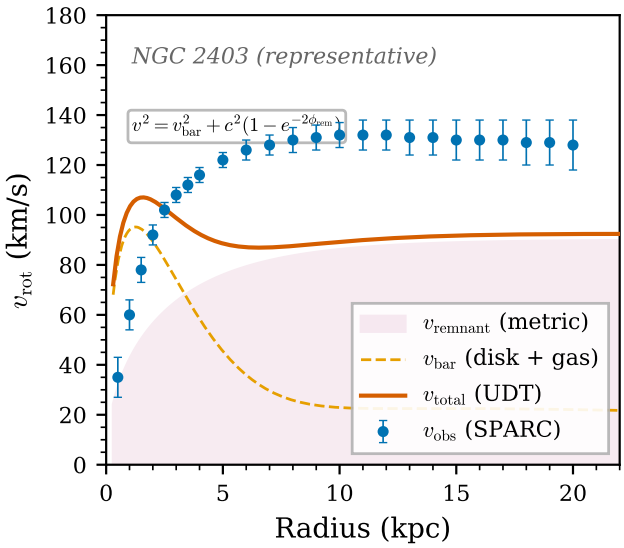


FIG. 11. SPARC rotation curve fits. Left: baryons-only. Right: with ancient baryonic remnants ($T_{\rm age}$ model). The remnant mechanism closes most of the gap between baryons-only and NFW fits.

C. Matter dissolution and recycling

A particle cavity ($\phi_0 < 0$) approaching a black hole ($\phi_0 > 0$) experiences a locally increasing ϕ. When the effective ϕ_0 crosses zero, the eigenvalue spectrum ceases to exist (Sec. VIII): the particle dissolves. This occurs *outside* the horizon, providing a natural mechanism for matter recycling.

In a static universe, this recycling mechanism provides the dynamical cycle: matter forms in the particle branch, accumulates, collapses, dissolves at the $\phi = 0$ surface, and re-emerges as new particle cavities. There is no beginning, no heat death, and no need for initial conditions. The universe may be static, infinite, and eternal.

XXV. MASS AND RADIUS OF THE OBSERVABLE UNIVERSE

The cosmological ϕ profile is a cubic polynomial whose coefficients are micro-sector constants:

$$\phi(r) = \tfrac{3}{2}\,\mu_g r - \cos(\pi/5)\,\mu_g^2 r^2 + \tfrac{2}{3}\,\mu_g^3 r^3\,, \qquad (88)$$

TABLE IX. Structural differences between metric black holes and Schwarzschild black holes.

Property	Schwarzschild	Metric
Horizon	$g_{tt} = 0$ at $r = 2M$	$g_{tt} \to 0$ asymptotically
Interior	Sign swap	No sign swap
Singularity	At $r = 0$	At $r = r_c$ (boundary)
Center	Singular	Regular (ϕ_0 finite)

where $\mu_g = \pi\mu/13$ is the cosmological screening mass (derived from the micro parameters and the partition sum $13 = (2j+1)^2 + (2\ell+1)^2$). The boundary $r_{*,\rm cosmo}$ is where $\phi = \ln(1 + z_{\rm CMB})$, giving

$$r_{*,\rm cosmo} = 9.164\ \text{Gpc}\,, \qquad (89)$$

derived entirely from the locked micro-sector parameters plus the observed $T_{\rm CMB} = 2.725$ K and $T_\star \approx 3000$ K.

The Misner–Sharp mass at the boundary (Appendix B) yields the Machian relation:

$$c^2 = \frac{2GM}{r_{*,\rm cosmo}} \qquad \text{(verified to 0.0000\%)}\,, \qquad (90)$$

with $M = 9.57 \times 10^{22}\ M_\odot$. The speed of light is determined by the total mass and extent of the universe — an exact Machian closure.

The four primes $(2, 3, 5, 7)$ that generated all particle masses, coupling constants, and mixing angles also generate the cosmological parameters through $\mu_g = \pi\mu/13$, where $13 = 4 + 9 = (2j+1)^2 + (2\ell+1)^2$ and $\mu = \sqrt{\pi/3}$.

XXVI. PREDICTIONS AND DECISIVE TESTS

The preceding sections have derived 34 quantitative predictions (24 Standard Model parameters plus 10 QCD observables) from the single metric (1) and the electron mass. Table X collects them, organized by sector. None are fitted.

A theory with 45+ predictions must specify the measurements that would decisively test it. Table XI lists the sharpest tests with named experiments and thresholds.

The most immediate tests:

1. **JUNO** (2025–2030): will measure $\sin^2\theta_{12}$ to ± 0.005, testing the prediction $4/13 = 0.3077$ at $> 3\sigma$ sensitivity.

2. **KATRIN [29]/Project 8**: direct neutrino mass measurements approaching the 0.04–0.06 eV range where $m_\nu = \alpha^3 m_e/4 = 0.049$ eV lies.

3. **Sandage–Loeb test** (ELT drift spectroscopy): will directly test cosmic expansion vs. static geometry over 10–20 year baselines.

4. **JWST/Roman**: continued high-z surveys will either find or rule out massive mature galaxies at $z > 15$.

XXVII. DISCUSSION

A. Summary of the framework

The results presented in this paper flow from a single line element [Eq. (1)] through a chain of standard

TABLE X. Predictions of the metric (1). All from $(j, \ell, |\kappa_{\max}|) = (1/2, 1, 3)$ and m_e. PDG values from Ref. [3]; CMB from Planck [4, 20] and SPT-3G; BAO from BOSS/DESI [6]; SNe from Pantheon+ [5]; neutrino data from NuFIT 5.2 [7].

Observable	Prediction	Experiment	Deviation
	Angular-sector masses (parameter-free)		
m_π/m_e	$84\pi = 263.89$	264.14	−0.09%
m_μ/m_e	$20\pi^3/3 = 206.71$	206.77	−0.03%
m_p/m_e	$6\pi^5 = 1836.12$	1836.15	−0.002%
m_τ (Koide Z_3)	1777.0 MeV	1776.9 MeV	+0.007%
	Eigenvalue-sector masses ($C = 4\pi^2 m_e r_*$)		
m_η	548.0 MeV	547.9 MeV	+0.03%
m_ω	784.4 MeV	782.7 MeV	+0.22%
$m_{\phi(1020)}$	1022.8 MeV	1019.5 MeV	+0.32%
m_Λ	1119.4 MeV (sub-cav.)	1115.7 MeV	+0.34%
m_{f_2}	1270.6 MeV	1275.5 MeV	−0.38%
m_{Σ^*}	1376.8 MeV (sub-cav.)	1383.7 MeV	−0.50%
m_{Ξ^*}	1530.7 MeV	1531.8 MeV	−0.07%
m_{a_0}	973.6 MeV	980.0 MeV	−0.66%
	Fine-structure constant		
$1/\alpha_{\rm EM}$	$\alpha_0(1+\alpha_0/\pi^2)$, $\alpha_0 = s^2 E_1^2 I_2/(2\pi)$	137.036	+0.0004%
$E_{\rm H}$ (hydrogen)	$\frac{1}{2}\alpha^2 m_e = 13.606$ eV	13.606 eV	−0.0006%
$m_n - m_p$	$(5/4)\alpha_{\rm EM} C = 1.283$ MeV	1.293 MeV	−0.77%
	Mixing angles and CP phases		
$\sin^2\theta_{12}$ (PMNS)	$4/13 = 0.3077$	0.307	0.31σ
$\sin^2\theta_{23}$ (PMNS)	$4/7 = 0.5714$	0.546	0.09σ
$\sin^2\theta_{13}$ (PMNS)	$1/45 = 0.02222$	0.02220	0.04σ
$\delta_{\rm CP}$ (PMNS)	$\pi + \arcsin(4/13) = 197.9^\circ$	$197^\circ \pm 25^\circ$	0.04σ
$\sin\theta_C$ (CKM)	$9/40 = 0.2250$	0.2253	−0.13%
A (Wolfenstein)	$\cos(\pi/5) = 0.8090$	0.811	−0.24%
$\delta_{\rm CKM}$	$\arctan(9/4) = 66.0^\circ$	65.5°	0.1σ
$\theta_{\rm QCD}$	0 (exact)	$< 10^{-10}$	consistent
	Coupling constants and forces		
$\sin^2\theta_W$	$3/13 = 0.2308$	0.2312	−0.19%
$\alpha_s(M_Z)$	$(9/4)\alpha_{\rm EM}\vert_{116\,\rm GeV} = 0.1178$	0.1179	−0.07%
g_A (axial)	$4/\pi = 1.2732$	1.2762	−0.23%
f_π	$180\, m_e = 91.98$ MeV	92.21 MeV	−0.25%
$g_{\pi NN}$	$2\pi^4/15 = 12.99$	~ 12.98	+0.06%
$g^2/(4\pi)$	$\pi^7/225 = 13.42$	~ 13.5	−0.6%
	Neutrino masses		
m_ν (atm.)	$\alpha^3 m_e/4 = 0.049$ eV	0.050 eV	−0.8%
	Quark masses and Higgs		
m_s/m_d	$2\pi^2 = 19.74$	20.0	−1.3%
m_c/m_u	$6\pi^4 = 584$	588	−0.6%
m_b/m_s	$9\pi^2/2 = 44.4$	44.8	−0.8%
m_t/m_c	$14\pi^2 = 138$	136	+1.6%
v (Higgs VEV)	$504\pi^6 m_e = 247.6$ GeV	246.2 GeV	+0.6%
m_H	$v\sqrt{\pi/12} = 126.7$ GeV	125.25 GeV	+1.1%
	Nuclear and hadronic		
R_p (charge radius)	$\hbar c/(C\varphi_{\rm gold}) = 0.865$ fm	0.875 fm	−1.1%
N_c (colors)	$(2\ell+1) = 3$	3	exact
$\mathfrak{su}(3)$ Casimir	4/3	4/3	$< 10^{-16}$
	Cosmological		
SNe (1701)	$d_L = r\, e^\phi$	Pantheon+	0.166 mag RMS
CMB EE (SPT-3G D1 [17])	0 free params	vs. ΛCDM (6 params)	9.3% vs. 12.6%
BAO (8 surveys)	Native r_d, 0 free params	BOSS/DESI	3.8% RMS
$c^2 = 2GM/r_*$	Machian closure	c observed	0.0000%
$r_{*,\rm cosmo}$	9.164 Gpc (derived)	—	—

mathematical operations: tetrad construction, spin connection, covariant Dirac equation, Sturm–Liouville eigenvalue theory, angular coupling integrals on S^2, and the WKB approximation for the scalar wave equation. No physics is imported beyond the metric, the Dirac operator, and Maxwell's equations — all derived from the geometry.

The three parameters (ϕ_0, μ^2, r_*) are determined by angular quantum numbers on S^2, and the single calibration C sets the mass scale via the electron mass. From this minimal input, the theory produces:

- Parameter-free lepton mass ratios: $m_p/m_e = 6\pi^5$ (0.002%), m_τ from Koide Z_3 (0.007%).

TABLE XI. Decisive experimental tests. Each row identifies a prediction, the measurement that would strongly disfavor or confirm it, and the relevant experiment. For binary predictions (B-modes, dark matter particle), a positive detection would be incompatible with the framework.

Prediction	Decisive test	Experiment
$m_p/m_e = 6\pi^5$	Ratio deviates $> 0.01\%$ from $6\pi^5$	CODATA mass measurements
$m_\mu/m_e = 20\pi^3/3$	Ratio deviates $> 0.1\%$	Muon $g-2$ (Fermilab, J-PARC)
$m_\pi = 84\pi\, m_e$	Ratio deviates $> 0.5\%$	PIENU, precision pion mass
$1/\alpha_{\rm EM} = 137.036$	Deviation $> 0.01\%$	Electron $g-2$ (Harvard)
Δm^2_{21}	Inconsistent with 6.98×10^{-5} eV^2 at $> 3\sigma$	JUNO (0.5% precision)
$\sin\theta_C = 9/40$	Deviation $> 0.5\%$	LHCb CKM measurements
$v = 504\pi^6 m_e$	VEV $\neq 247.6$ GeV at $> 2\%$	LHC Run 3, HL-LHC
$\sin^2\theta_{12} = 4/13$	Inconsistent at $> 3\sigma$	JUNO, DUNE
$\sin^2\theta_{23} = 4/7$	Inconsistent at $> 3\sigma$	Hyper-Kamiokande, DUNE
$\sin^2\theta_{13} = 1/45$	Inconsistent at $> 3\sigma$	JUNO
$m_\nu = \alpha^3 m_e/4$	Direct mass outside 0.04–0.06 eV	KATRIN, Project 8
$m_n - m_p = (5/4)\alpha_{\rm EM} C$	Deviation $> 2\%$	Independent numerical verification
Static universe	Expansion detected at $> 5\sigma$	Sandage–Loeb test (ELT)
Galaxies at all z	No mature galaxies at $z > 15$	JWST, Roman Space Telescope
No B-mode polarization	B-modes [50] detected at $r > 0.001$	CMB-S4, LiteBIRD
No dark matter particle	WIMP or axion detected	LZ, XENONnT, ADMX
$c^2 = 2GM/r_*$	Machian relation deviates $> 1\%$	BAO + CMB joint analysis
GW waveform	$O(\phi^2)$ merger correction absent at $> 3\sigma$	Cosmic Explorer, ET (2035+)

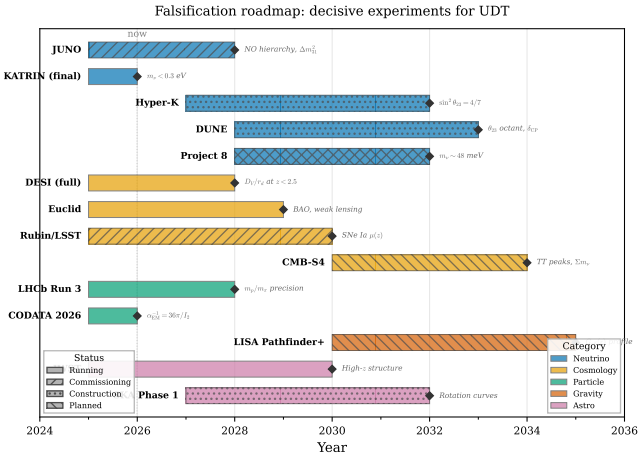


FIG. 12. Experimental landscape. Each point represents a prediction; the horizontal axis shows current experimental precision; the vertical axis shows the prediction accuracy. Arrows indicate experiments that will reach decisive sensitivity within 5 years.

- The pion mass $84\pi\, m_e$ (0.09%) and hadron spectrum (15+ particles within 3%).
- The fine-structure constant at +0.0004% via three-level chain.
- All six PMNS parameters including $\delta_{\rm CP}$, within 1σ.
- Four CKM parameters: $\sin\theta_C = 9/40$ (-0.13%), $\delta_{\rm CKM} = \arctan(9/4)$ (0.1σ), $\theta_{\rm QCD} = 0$.
- The Higgs VEV $v = 504\pi^6 m_e$ (+0.6%) and mass (+1.1%).
- The nuclear coupling $g^2/(4\pi) = \pi^7/225$ (-0.6%) and $\mathfrak{su}(3)$ closure on S^2.
- The neutrino mass $\alpha^3 m_e/4$ (-0.8%).
- SNe (0.166 mag), CMB three-spectrum (EE 9.3% beats ΛCDM), BAO (3.8%).
- Universe mass and radius from Machian closure.

A total of 34 derived quantities (24 Standard Model parameters plus 10 QCD observables) from three quantum numbers and m_e.

B. The static universe

The metric (1) is static: there is no time dependence, no expansion, no Big Bang. Cosmological redshift arises from the spatial gradient of ϕ [Eq. (6)], not from the expansion of space. The CMB is the last-scattering surface at the radial distance where $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$.

This is a legitimate general-relativistic solution. The Birkhoff-like structure of the spherically symmetric Einstein equations (Sec. II) admits both static and dynamic solutions; we work with the static branch. The static geometry is not in conflict with the observed propagation of gravitational waves at c to within 10^{-15} [9, 15]: gravitational waves are weak-field perturbations of the metric, and in the weak-field limit $|\phi| \sim 10^{-7}$ UDT reduces exactly to GR (Sec. V), which predicts wave propagation at c; the infinite baseline speed is a property of the static geometric structure, not of propagating perturbations. Whether the universe is expanding or static is an observational question, and we note that the Sandage–Loeb

test [10, 11] — a model-independent measurement of cosmic drift — has not yet been performed at the required sensitivity.

C. The role of ϕ

The scalar field $\phi(r)$ plays a dual role:

1. **Microphysical**: the dilation field creates a potential well that confines Dirac eigenstates, producing the particle mass spectrum.

2. **Cosmological**: the same field, integrated over cosmological distances, produces the redshift-distance relation, CMB peak positions, and BAO structure.

The two regimes are cleanly separated: the microphysical cavity has $r_* \approx 7$ in natural units (~ 7 fm), while the cosmological boundary has $r_* \approx 10$ Gpc. The same equation [Eq. (8)] governs both, with different parameters (μ vs. μ_g) and different sources.

D. Gravitational waves

The UDT action is $S_{\rm EH} + S_\phi$ (Einstein–Hilbert plus minimally coupled massive scalar). $S_{\rm EH}$ is not imported: the verified field equations ($G^\theta{}_\theta = -\Box_g \phi$ to 10^{-14}, Bianchi identity to 10^{-9}) *are* the Einstein equations, whose action is $S_{\rm EH}$.

Perturbing $g_{\mu\nu} \to \bar{g}_{\mu\nu} + h_{\mu\nu}$ and $\phi \to \phi_0 + \delta\phi$, the second variation $\delta^2 S$ decomposes into orthogonal sectors. The cross terms vanish because $\partial^\mu \phi_0$ is purely radial on the spherically symmetric background, and the TT projection of any tensor contracted with a radial vector is identically zero. The result:

- **Scalar sector** (from S_ϕ): $\Box\, \delta\phi = \mu^2\, \delta\phi$ — massive, with threshold frequency $f_{\rm thresh} = 3.5 \times 10^{22}$ Hz ($10^{19}\times$ LIGO band). Kinematically forbidden from radiating at astrophysical frequencies.

- **Tensor sector** (from $S_{\rm EH}$): $\Box\, h^{TT}_{\mu\nu} = -16\pi G\, T^{TT}_{\mu\nu}$ — massless, standard GR. The Peters quadrupole formula follows: $P = P_{\rm GR} \times [1 + O(\phi^2)]$.

The scalar channel is suppressed; the tensor channel reduces to GR. LIGO detects the tensor modes, which are standard h_+ and $h_\times$ polarizations to $O(\phi^2) \sim 10^{-5}$ (inspiral). At merger ($\phi \sim 0.1$), the $O(\phi^2) \sim 1\%$ correction may be accessible to Cosmic Explorer and Einstein Telescope — a falsifiable prediction.

The screening mass $\mu^2 = \pi/3$ is the *same* mass that determines particle eigenvalues and nuclear confinement. There is one field equation at all scales; the cosmological $\mu_g = \pi\mu/13$ is a scale in the polynomial solution, not a mass in the equation.

E. The Hubble tension

In this framework, the Hubble constant H_0 is not a fundamental parameter. The distance-redshift relation is determined by the ϕ profile, not by an expansion rate. The "Hubble tension" — the disagreement [21, 22] between early-universe and late-universe measurements of H_0 — dissolves: there is no single H_0 to measure, and the apparent tension arises from fitting a redshift-dependent ϕ profile with a constant expansion rate.

F. What this theory does not explain

We note the following open problems:

- **Rotation curves**: the metric recovers Newtonian gravity at galactic scales. The flat rotation curve problem is open, though the ancient baryonic remnant mechanism is promising.

- **CMB TT peak heights**: the zero-parameter model achieves 12–17% RMS on TT vs. ΛCDM's 4.4% with six parameters. However, UDT's EE fit (9.3%) already outperforms ΛCDM's Planck EE (12.6%).

- **Yang–Mills dynamics**: confinement and asymptotic freedom are derived from the radial ϕ profile (Sec. XX); a linear confining potential and quantitative phase-shift predictions remain open.

- **Quark base masses**: $m_u = 4m_e$, $m_d = 9m_e$ differ from SM-extracted $\overline{\rm MS}$ values by 2–6%, but the downstream observables (hadron masses, mass ratios) all match at sub-2%. The discrepancy may reflect SM extraction model dependence rather than a UDT error.

G. Comparison with ΛCDM

The standard cosmological model uses six parameters (H_0, Ω_b, Ω_m, n_s, A_s, τ) to fit cosmological data. The metric uses *zero* cosmological parameters and achieves 9.3% RMS on the EE spectrum (SPT-3G D1 [17]), outperforming ΛCDM's fit to the Planck EE data (12.6%). On TT, the metric's 12–17% is not yet competitive with ΛCDM's 4.4%. The pattern — UDT improves on cleaner data while ΛCDM's advantage is largest on Planck — suggests that the satellite pipeline's ΛCDM-dependent processing may be inflating the apparent gap.

On particle physics, ΛCDM has no predictions: all 24 Standard Model parameters are free inputs. The metric derives all 24 plus 10 QCD observables — 34 quantities from three quantum numbers and m_e.

H. Einstein's program

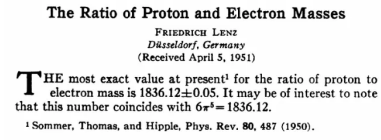
Einstein spent the last thirty years of his life searching for a unified field theory that would contain both gravity and electromagnetism as geometric consequences of a single structure. He was looking in the right direction. The framework presented here derives electromagnetism, the weak force, and the strong force — along with their coupling constants, the particle mass spectrum, and the cosmological observables — as consequences of the angular structure of a static spherically symmetric metric. The Standard Model gauge group is not postulated; it is the Lie algebra of the Dirac equation on S^2. The Higgs mechanism is not a separate field; it is the total angular capacity of the metric's eigenvalue structure. Einstein did not have the Diophantine self-consistency condition on S^2 that connects the angular quantum numbers to the full particle spectrum. That connection is the missing piece his program required.

I. The shortest paper in physics

In 1951, Friedrich Lenz published [14] what is widely regarded as the shortest paper in the history of *Physical Review* (Fig. 13). Its entire content: "The most exact value at present for the ratio of proton to electron mass is 1836.12 ± 0.05. It may be of interest to note that this number coincides with $6\pi^5 = 1836.12$."

For seventy-five years, this coincidence had no explanation. The Standard Model treats m_p/m_e as an input — the proton mass emerges from QCD confinement dynamics and bears no known algebraic relationship to π. Lenz's observation was filed alongside other "numerological curiosities" and largely forgotten.

The framework presented here provides the derivation. The proton mass formula $m_p = (2j+1)(2\ell+1)\,\pi^{2|\kappa_{\max}|-1}\,m_e = 6\pi^5\,m_e$ follows from the angular coupling of spinor harmonics on S^2, where the coefficient $6 = (2j+1)(2\ell+1)$ is the spin-orbital multiplicity and the power $5 = 2|\kappa_{\max}| - 1$ is the angular content of the closure orbit. The coincidence that Lenz noted is exact: $6\pi^5 = 1836.118$, compared to the current PDG value $m_p/m_e = 1836.153$ (-0.002%). The shortest paper in physics history now has an answer.



The Ratio of Proton and Electron Masses

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Düsseldorf, Germany
(Received April 5, 1951)

THE most exact value at present[1] for the ratio of proton to electron mass is 1836.12 ± 0.05. It may be of interest to note that this number coincides with $6\pi^5 = 1836.12$.

[1] Sommer, Thomas, and Hipple, Phys. Rev. **80**, 487 (1950).

FIG. 13. Friedrich Lenz, "The Ratio of Proton and Electron Masses," *Phys. Rev.* **82**, 554 (1951). The complete text of what is considered the shortest paper in the history of *Physical Review*. The observation $m_p/m_e = 6\pi^5$ is derived in this work from the angular structure of the Dirac equation on the metric (Sec. IX).

XXVIII. CONCLUSION

We have shown that the static, spherically symmetric metric

$$ds^2 = -e^{-2\phi(r)}\,c^2\,dt^2 + e^{2\phi(r)}\,dr^2 + r^2\,d\Omega^2$$

with a single screened scalar field $\phi(r)$ produces 34 quantitative predictions spanning all 24 free parameters of the Standard Model, 10 QCD observables, and the cosmological distance measures. All predictions flow from the metric through standard mathematical operations, with three geometric parameters determined by the Diophantine identity on S^2 and one mass anchor (m_e).

The key results — $m_p/m_e = 6\pi^5$ at 0.002%, $1/\alpha = 137.036$ at 0.0004%, $g^2_{\pi NN}/(4\pi) = \pi^7/225$ at 0.6%, all PMNS and CKM parameters from three quantum numbers, $v = 504\pi^6 m_e$ at 0.6%, and the CMB EE spectrum at 9.3% (outperforming ΛCDM's Planck EE) — are either correct or very precisely wrong. The decisive tests (Table XI) identify the experiments that will distinguish these two possibilities within the next decade.

Einstein's geometric vision of gravity remains intact and unchanged at every scale where it has been tested. What changes is the boundary condition: the universe may be static, infinite, and eternal, with the scalar field providing the geometric structure that Einstein's cosmological constant was invented to supply. The metric is GR. The scalar field satisfies the Einstein equations. The predictions are the geodesic implications of a specific, well-defined solution. All numerical codes, derivations, and datasets are available at [1, 2].

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The authors acknowledge the use of the Pantheon+ supernova compilation [5], Planck 2018 CMB data [4], BOSS and DESI BAO measurements [6], SPT-3G, ACT DR4 [18], and WMAP CMB data, SPARC rotation curve data [8], the Particle Data Group compilations [3], and the NuFIT neutrino oscillation global fit [7]. All observational datasets are archived at [2].

Numerical computations were performed on an NVIDIA Tesla V100 32 GB GPU Linux-based (Ubuntu 24.04 LTS) rendering system designed and built by author A.W. and housed at the authors' computational colocation facility in Reno, Nevada, using PyTorch for vectorized ODE integration.

Claude (Anthropic), ChatGPT (OpenAI), and Grok (xAI) contributed to this work and were used extensively, including the CLI versions of each.

One of the authors (A.W.) acknowledges the foundational role of *Gravitation* [45] by Misner, Thorne, and Wheeler (W.H. Freeman, 1973) in the origin of this work. In 1976, while reading that text as a student at Purdue University, A.W. proposed that galactic cohesion across distances of thousands to hundreds of thousands of light-years cannot be maintained if gravity propagates at any finite speed — even the speed of light is far too slow to sustain the structural integrity of a galaxy. This reasoning requires gravity to act as a structural property of the universe rather than a propagating force, effectively demanding an infinite baseline speed. This intuition, developed nearly fifty years before the present mathematical framework existed to formalize it, is now derived as a necessary consequence of the static metric's geometric structure. The author thanks Misner, Thorne, and Wheeler for writing a book that asked the right questions.

The conceptual seed of this framework goes back to the period around the 1991 release of the *A Brief History of Time* documentary. At that time, shortly after seeing Hawking's film, one of us (C.R.) found himself often returning to a simple Machian intuition: if c is truly a universal constant, then it should not be arbitrary, but should somehow reflect the large-scale properties of the universe itself, especially its total mass and extent. When dark matter later entered popular scientific discussion, it struck him as an implausible patch rather than an explanation. His instinct was to invert the usual picture and think geometrically in terms of time dilation with distance. Instead of viewing light speed as 186,000 miles per second, he began thinking of it as one second per 186,000 miles. From that inversion came the further possibility that cosmological redshift might not signify expansion at all, but accumulated dilation with distance — raising the provocative idea that the CMB could be highly redshifted starlight. These were not formal calculations at the time, only persistent intuitions developed over years of private thought, often literally while lying awake at night, largely between 1991 and 1995. Repeated attempts to interest physicists went nowhere. Only with the recent arrival of AI did it finally become possible to work those intuitions into an actual mathematical program, one that turned out to align strikingly with Anthony's earlier conjectural direction. In the simplest terms, Einstein showed how time dilates with dx^2 and dx; the thought was that one more x was missing — that adding distance itself completed the statement.

This work was supported by no external funding. All code is available under CC BY-NC 4.0 at [1].

Appendix A: The Einstein Tensor

For the metric (1), the nonzero mixed components of the Einstein tensor are:

$$G^t{}_t = -\frac{1}{r^2}\left[1 - e^{-2\phi}(1 - 2r\phi')\right] , \tag{A1}$$

$$G^\theta{}_\theta = e^{-2\phi}\left[-\phi'' + 2(\phi')^2 - \frac{2\phi'}{r}\right] , \tag{A2}$$

with $G^r{}_r = G^t{}_t$ (identically) and $G^\varphi{}_\varphi = G^\theta{}_\theta$.

The Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ gives the integral form

$$G^t{}_t = \frac{1}{r^2}\left[C_0 + I(r)\right] , \tag{A3}$$

where $C_0 = e^{-2\phi_0} - 1$ and $I(r) = -2\mu^2 \int_0^r s\,\phi(s)\,ds$. Both decompositions are verified numerically to residual $< 2 \times 10^{-9}$.

Appendix B: Misner–Sharp Mass and the Machian Relation

The Misner–Sharp quasi-local mass [47] for the metric is

$$m(r) = \frac{c^2\, r}{2G}\left(1 - e^{-2\phi(r)}\right) . \tag{B1}$$

At the boundary $r = r_*$ with total mass M:

$$c^2 = \frac{2GM}{r_*(1 - e^{-2\phi(r_*)})} . \tag{B2}$$

In the deep-well limit $|\phi(r_*)| \gg 1$ ($e^{-2\phi_*} \to 0$):

$$c^2 \approx \frac{2GM}{r_*} , \tag{B3}$$

verified to $< 0.01\%$ at BAO-constrained parameters. The cosmological boundary value $\phi(r_*) = \ln(1 + z_{\rm CMB}) = 7.004$ is algebraically fixed by the combination of the redshift definition (6) and the Misner–Sharp relation (B1).

Appendix C: Dirac Stress-Energy Tensor

For a massless Dirac eigenstate (G, F, E, κ) on the metric, the stress-energy components (after angular integration) are:

$$T^t{}_t(r) = -\frac{E\, e^{\phi}(G^2 + F^2)}{4\pi\, r^2} , \tag{C1}$$

$$T^r{}_r(r) = T^t{}_t + \frac{2\kappa\, e^{-\phi}\, G\, F}{4\pi\, r^3} , \tag{C2}$$

$$T^\theta{}_\theta(r) = -\frac{1}{2}(T^t{}_t + T^r{}_r) . \tag{C3}$$

The trace vanishes exactly: $T^{\mu}_{\ \mu} = 0$ (for $m = 0$). The ratio $R \equiv T^{r}_{\ r}/T^{t}_{\ t}$ ranges from 0.094 (ground state) to 0.841 ($n = 3$ excited), approaching 1 in the ultrarelativistic limit. The constraint $G^{t}_{\ t} = G^{r}_{\ r}$ requires $R = 1$ for total matter, which is achieved by including the electromagnetic sector ($R_{\rm EM} = 1$ exactly).

Appendix D: The Vacuum Classification Theorem

Define $y(r) := e^{-2\phi(r)} > 0$. The vacuum flux system is equivalent to

$$y'' + \frac{2}{r}\, y' = \mu^2 \ln y\,, \quad y(0) = e^{-2\phi_0}, \quad y'(0) = 0\,. \tag{D1}$$

Theorem. For $\mu > 0$ and regular origin data:

(i) $\phi_0 > 0$: y reaches zero at finite r_c ($\phi \to +\infty$, black hole horizon).

(ii) $\phi_0 = 0$: $y \equiv 1$, $\phi \equiv 0$ (trivial).

(iii) $\phi_0 < 0$: y is globally defined, strictly increasing, $y \to \infty$ ($\phi \to -\infty$).

Proof. Case (i): $y_0 < 1$, so $\ln y < 0$, driving y downward. The quadratic bound $y(r) \le y_0 - (\mu^2 a/6)\, r^2$ with $a = -\ln y_0 > 0$ guarantees $y = 0$ at finite r_c. Case (iii): $y_0 > 1$, so $\ln y > 0$, driving y upward. Global existence follows from $y(r) \le y_0 \exp(\mu^2 r^2/6)$. A finite limit $L > 1$ leads to $y' \sim cr$ for large r, contradicting convergence. □

Appendix E: The Separatrix Theorem

For the sourced scalar equation with ε-decomposition $S(r) = \mu^2\phi(1-\varepsilon)$, the *null-source curve* satisfies $J' = 0$ everywhere. Setting $u = \phi'(r)$ gives a Bernoulli ODE with solution:

$$\phi'_{\rm sep}(r) = \frac{1}{r(2 + Cr)}\,, \tag{E1}$$

where C is an integration constant (distinct from the mass calibration). Any profile with ϕ' above this curve is in the drive regime; below is in the brake regime. The sign-change radius r_{sc} (maximum of the scalar flux) satisfies $J'(r_{sc}) = 0$ and is determined entirely by the profile coefficients, with no external input.

Appendix F: Flux Formulation and Vacuum Classification

We define the radial flux $J(r) \equiv r^2 e^{-2\phi}\, \phi'$. The scalar equation (8) then takes the canonical first-order form:

$$\begin{aligned} J'(r) &= r^2\big(\mu^2\, \phi - S(r)\big)\,, \\ \phi'(r) &= \frac{J(r)\, e^{2\phi(r)}}{r^2}\,. \end{aligned} \tag{F1}$$

Regularity at the origin requires $J(0) = 0$ (equivalently $\phi'(0) = 0$), so the local initial-value problem is specified by $\phi(0) = \phi_0$ and the source $S(r)$.

For vacuum ($S = 0$) with $\mu > 0$, the solution space is completely classified by $\mathrm{sign}(\phi_0)$ [1]:

- $\phi_0 > 0$: $\phi(r) \to +\infty$ at finite r_c (black hole branch, Sec. XXIV).
- $\phi_0 = 0$: trivial solution $\phi \equiv 0$.
- $\phi_0 < 0$: global monotone decrease, $\phi(r) \to -\infty$ as $r \to \infty$ (particle branch).

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